Transformer

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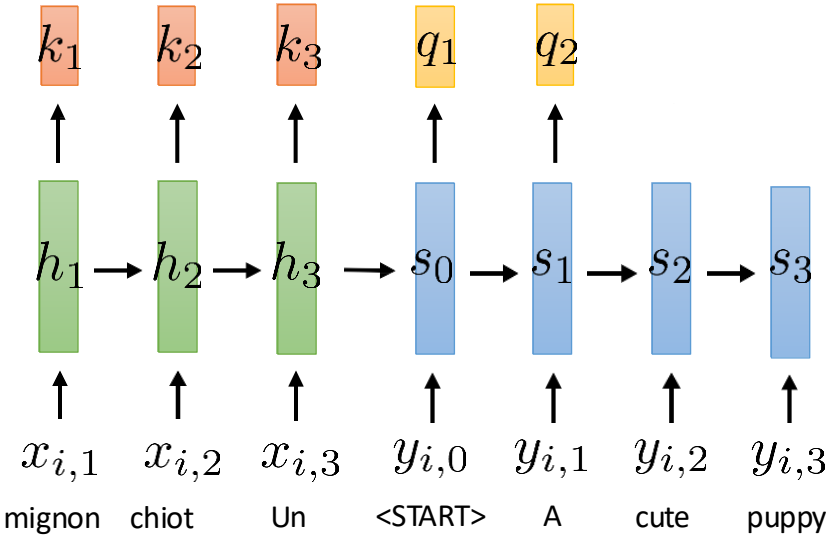
https://tanmoychak.com/

Slides are adopted from Sergey Levine

Is Attention All We Need?

Slides by Sergey Levine

Recap: Attention



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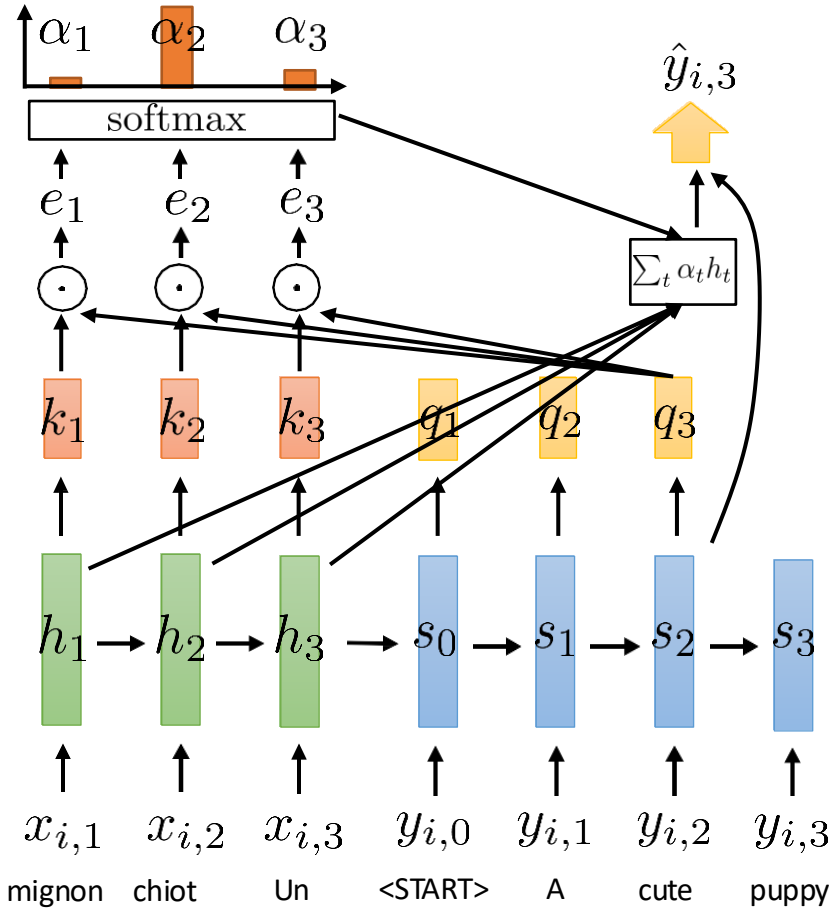

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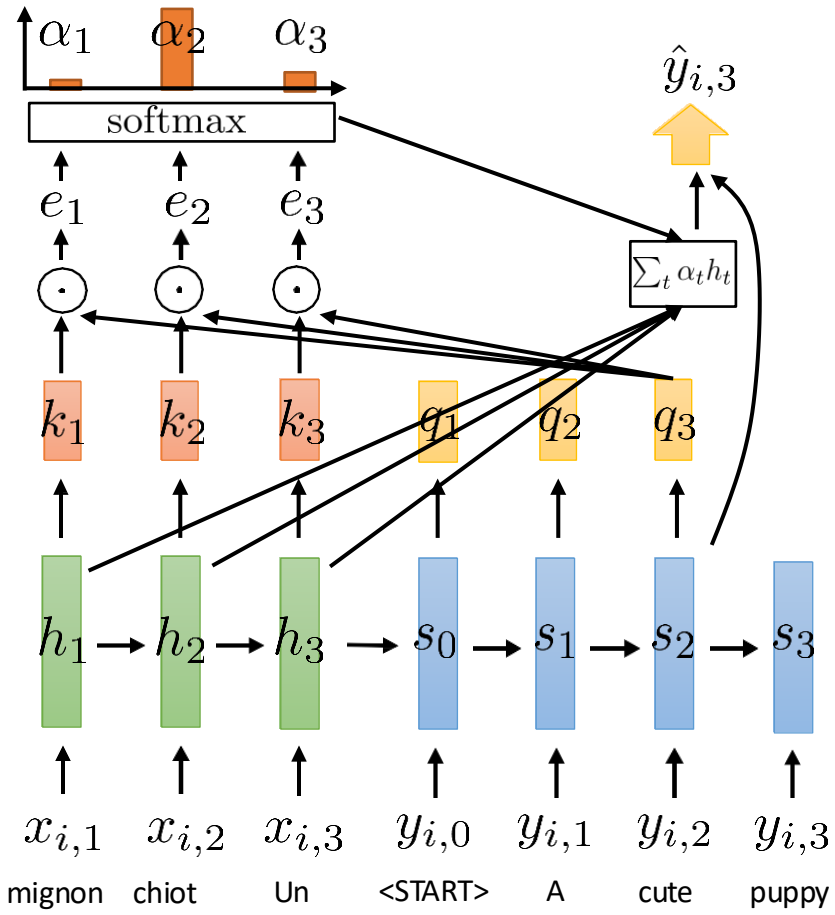

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- Can we transform our RNN into a **purely attention-based model**?
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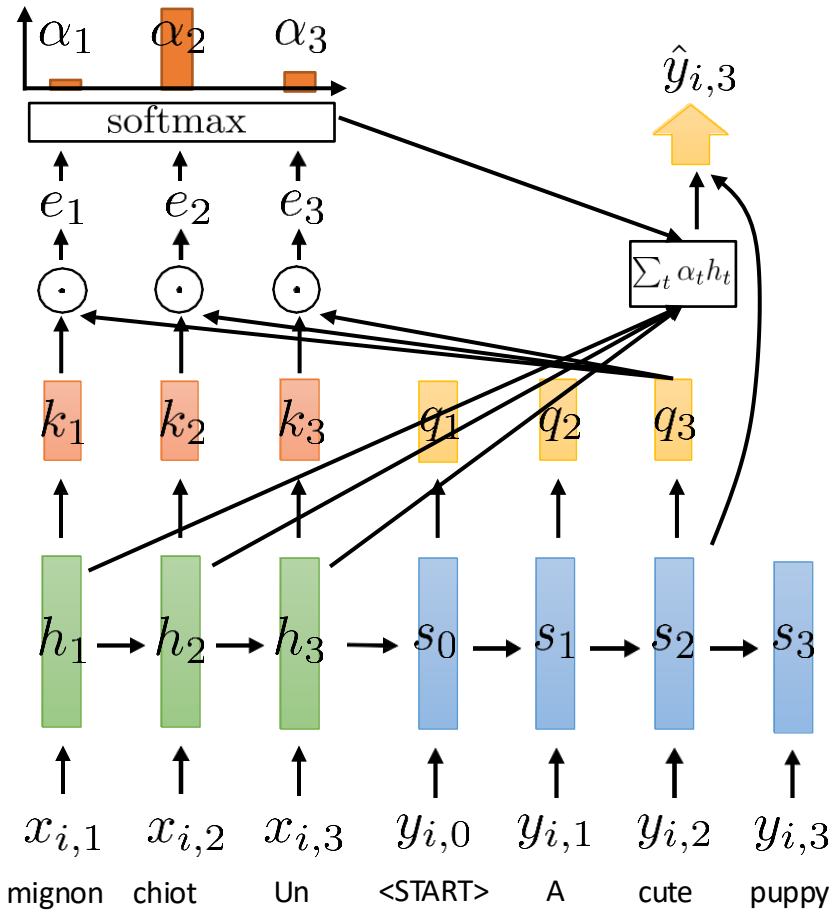

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The encoder lacks temporal dependencies at all!

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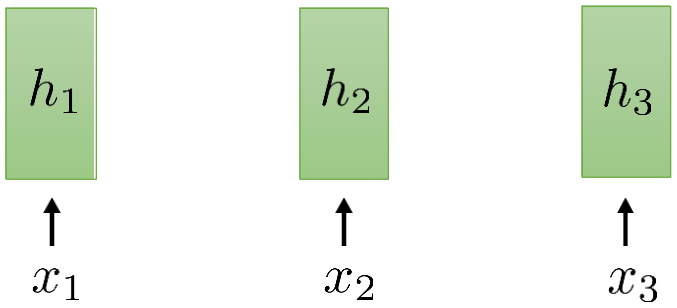

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but still weight sharing:

$$h_t = \sigma(W x_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

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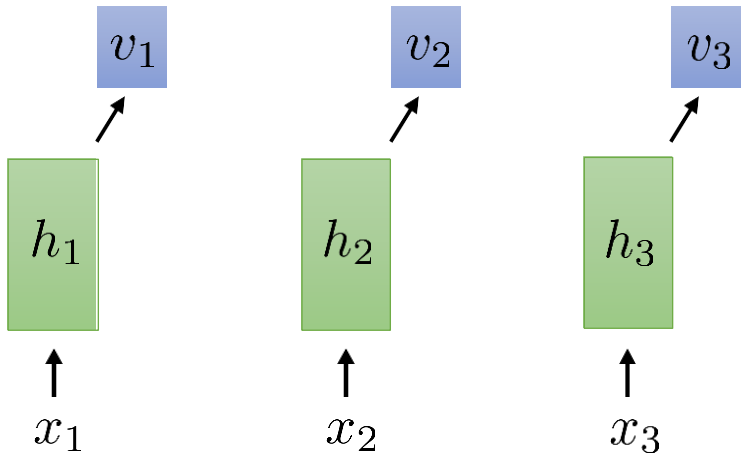

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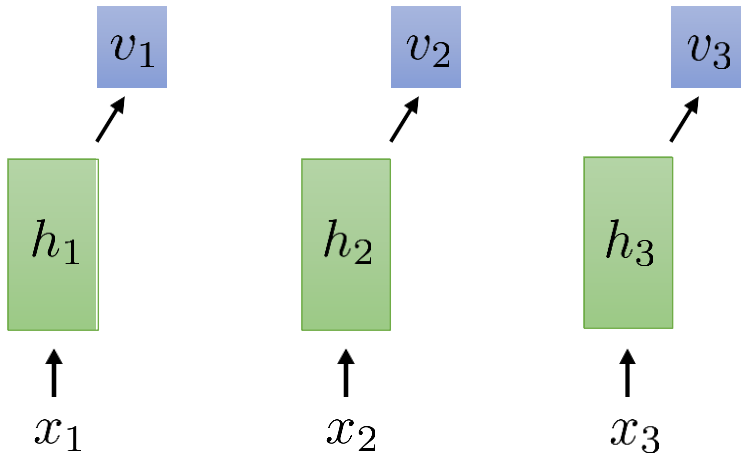

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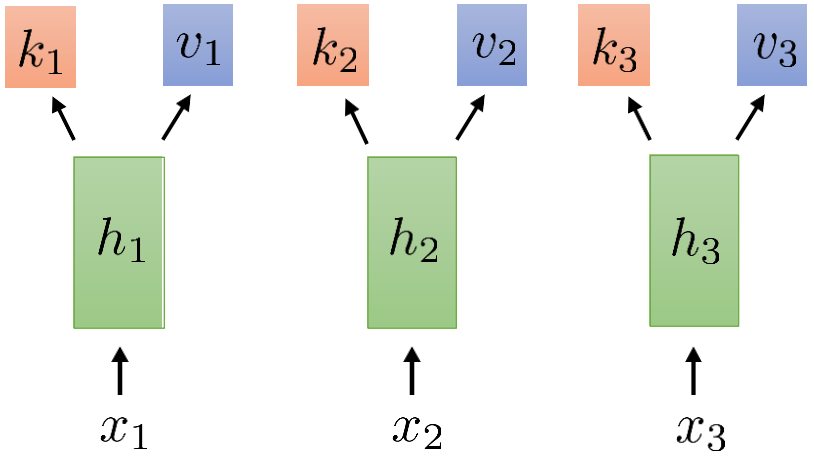

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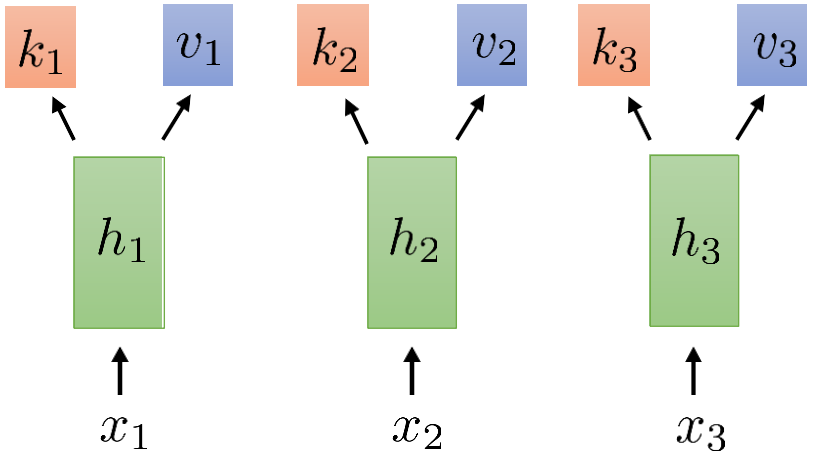

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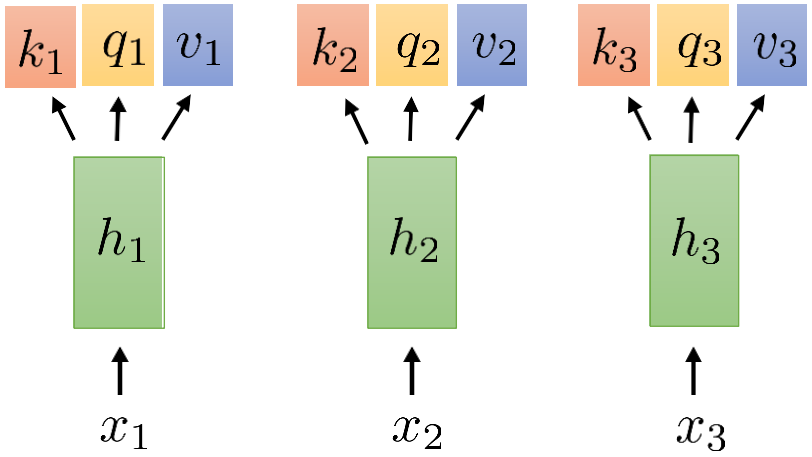

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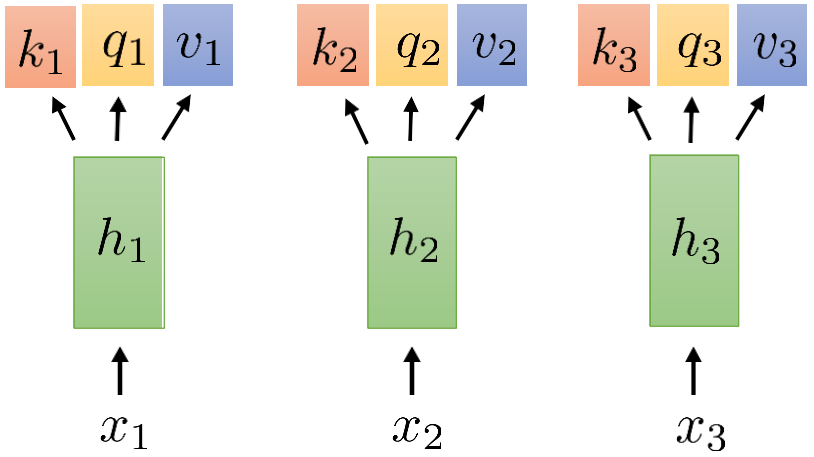

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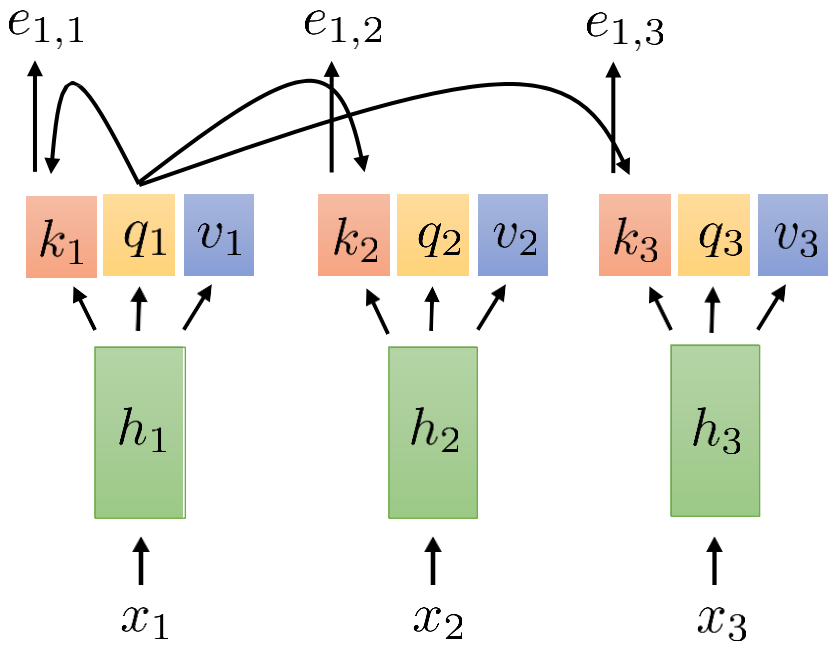

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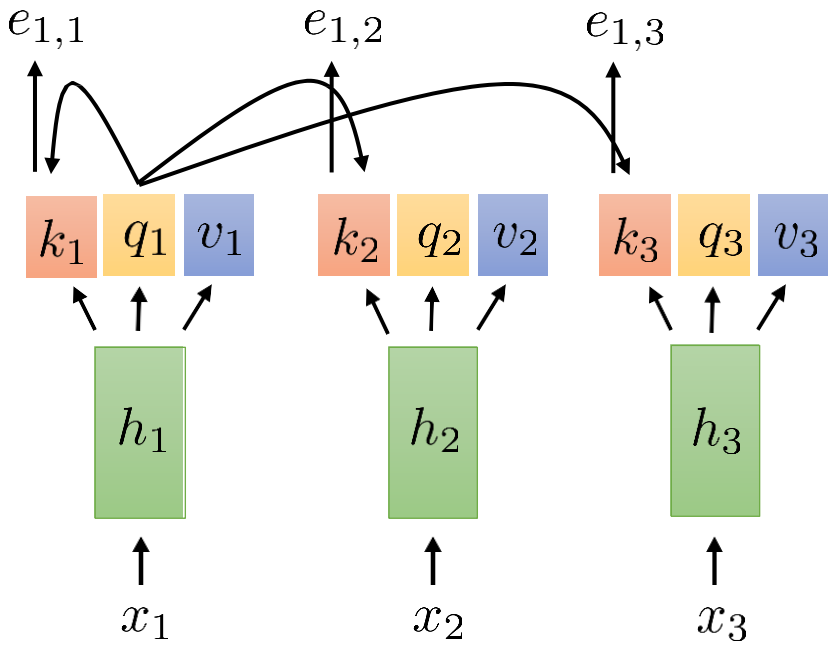

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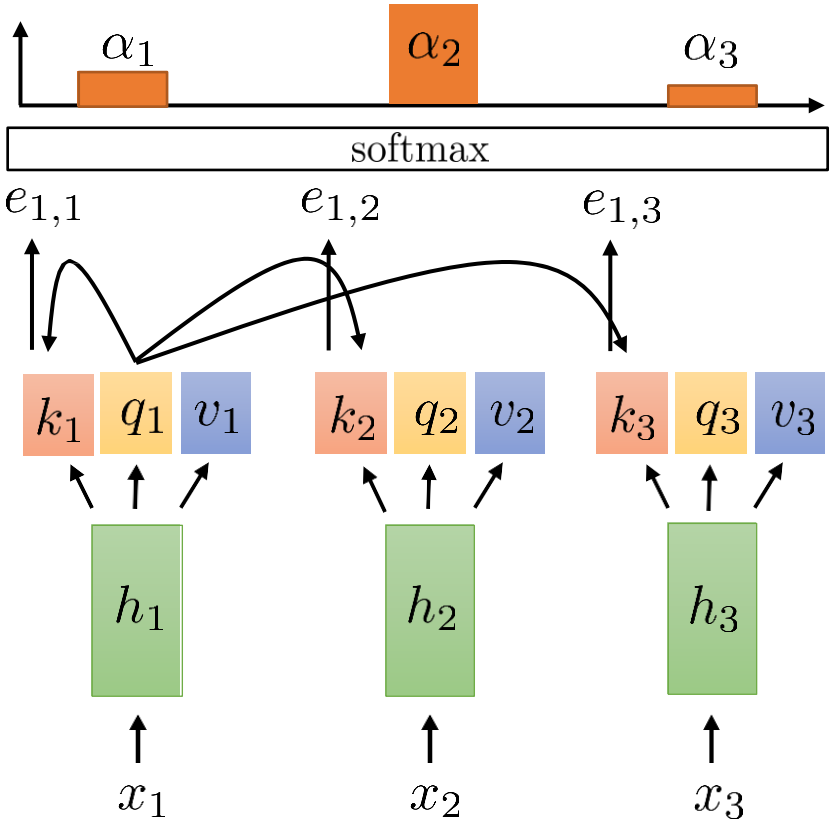

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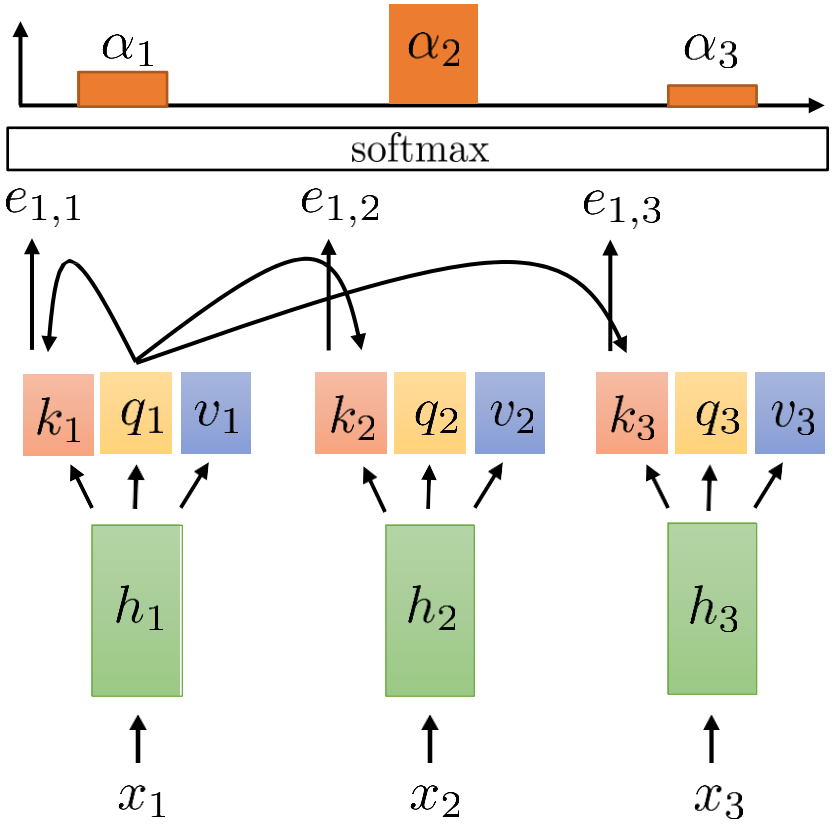

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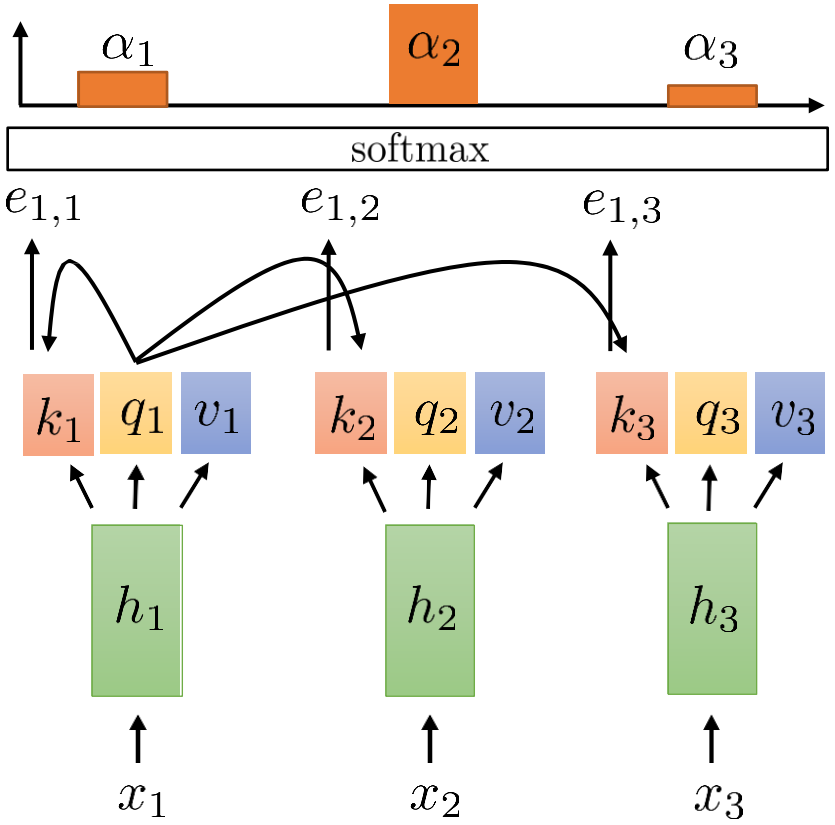

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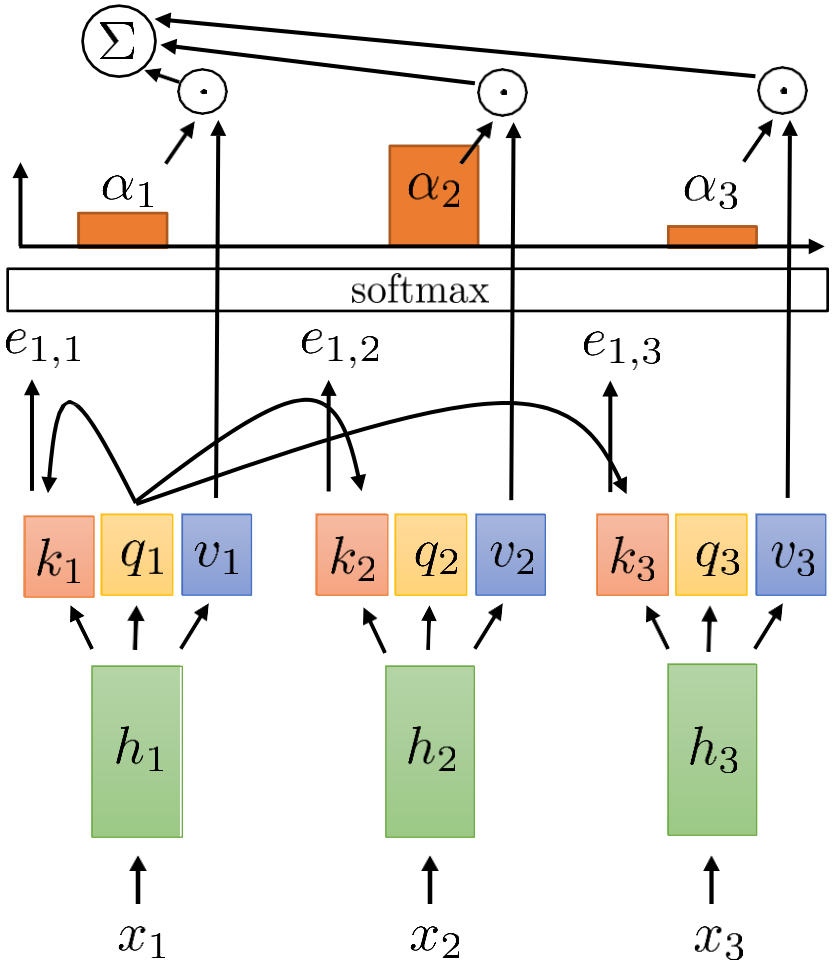

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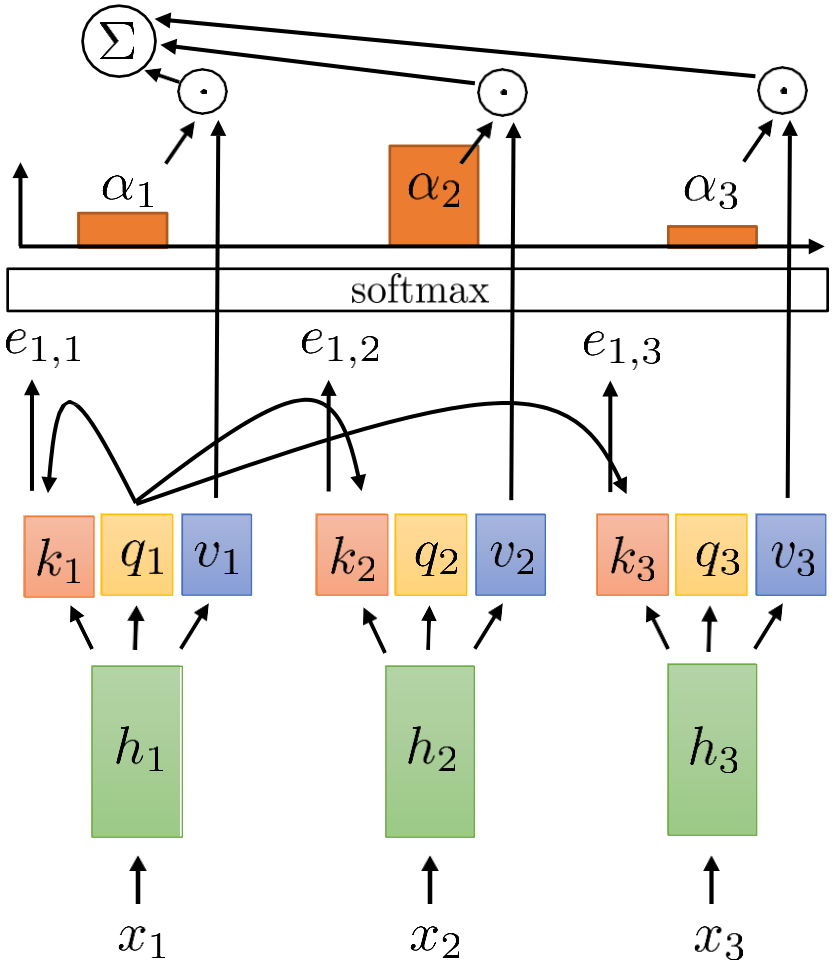

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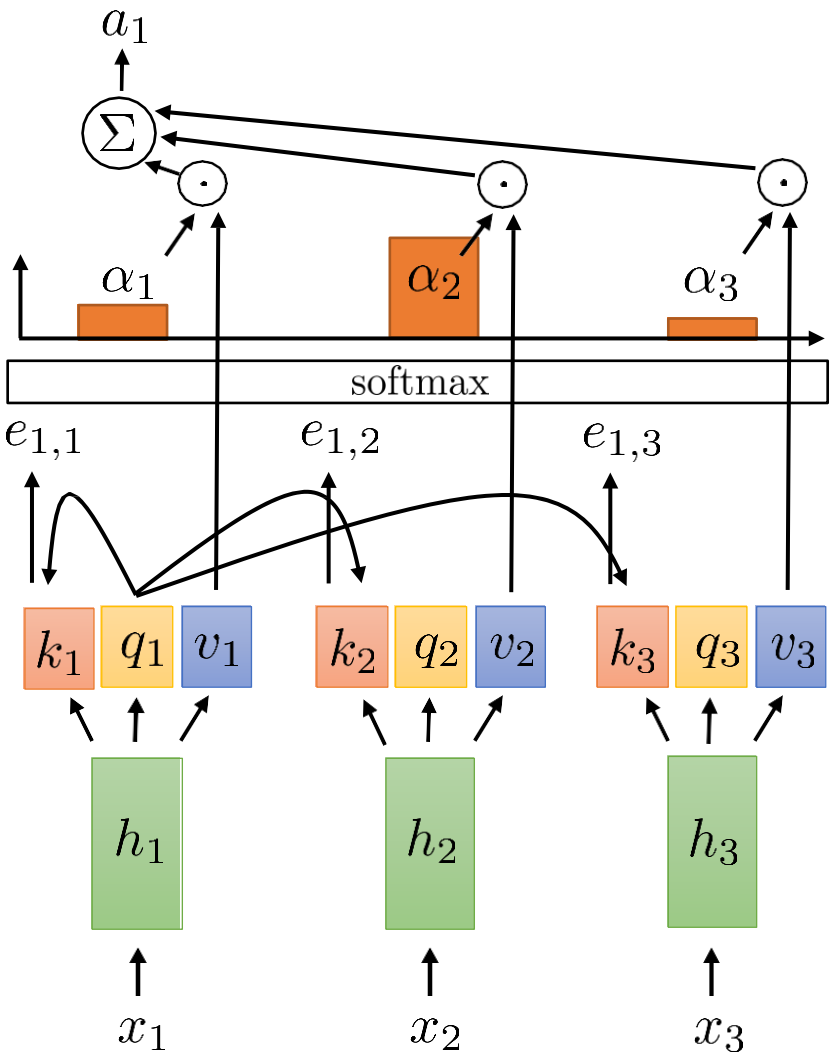

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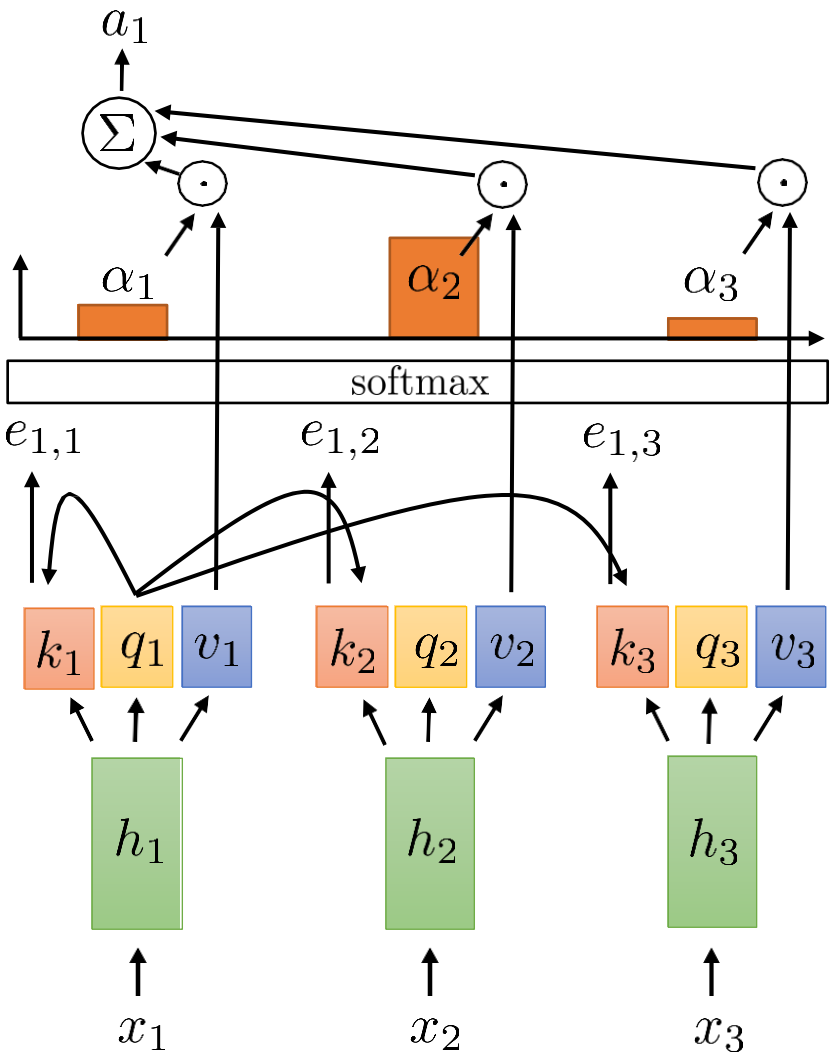

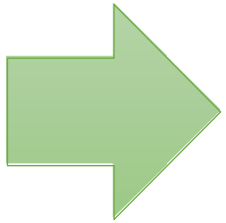

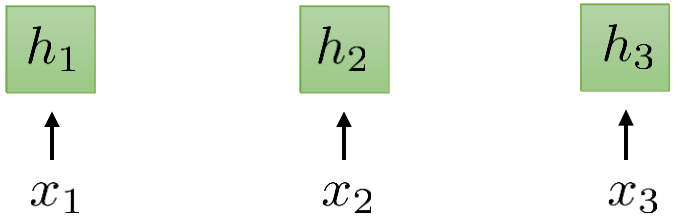

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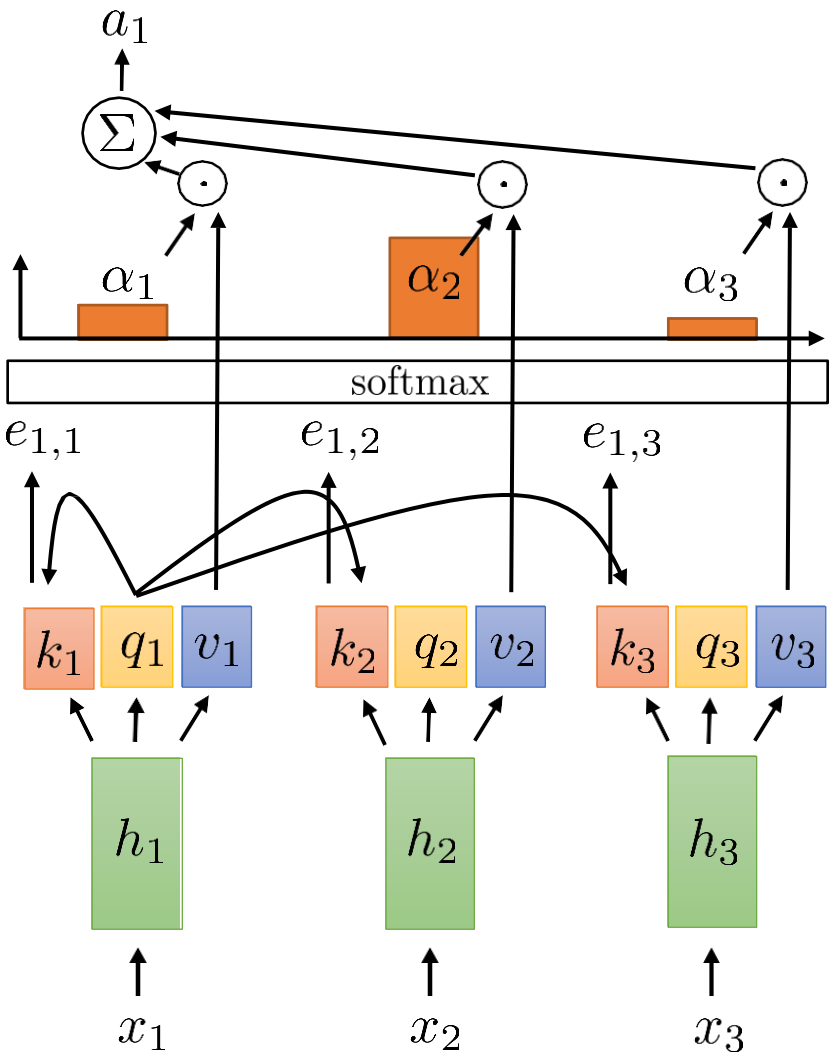

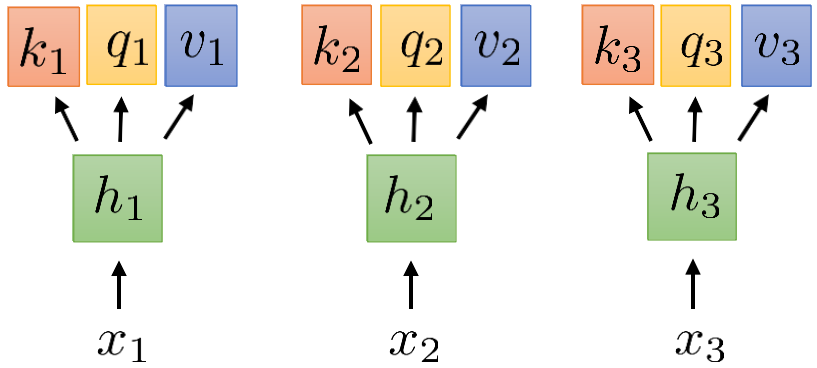

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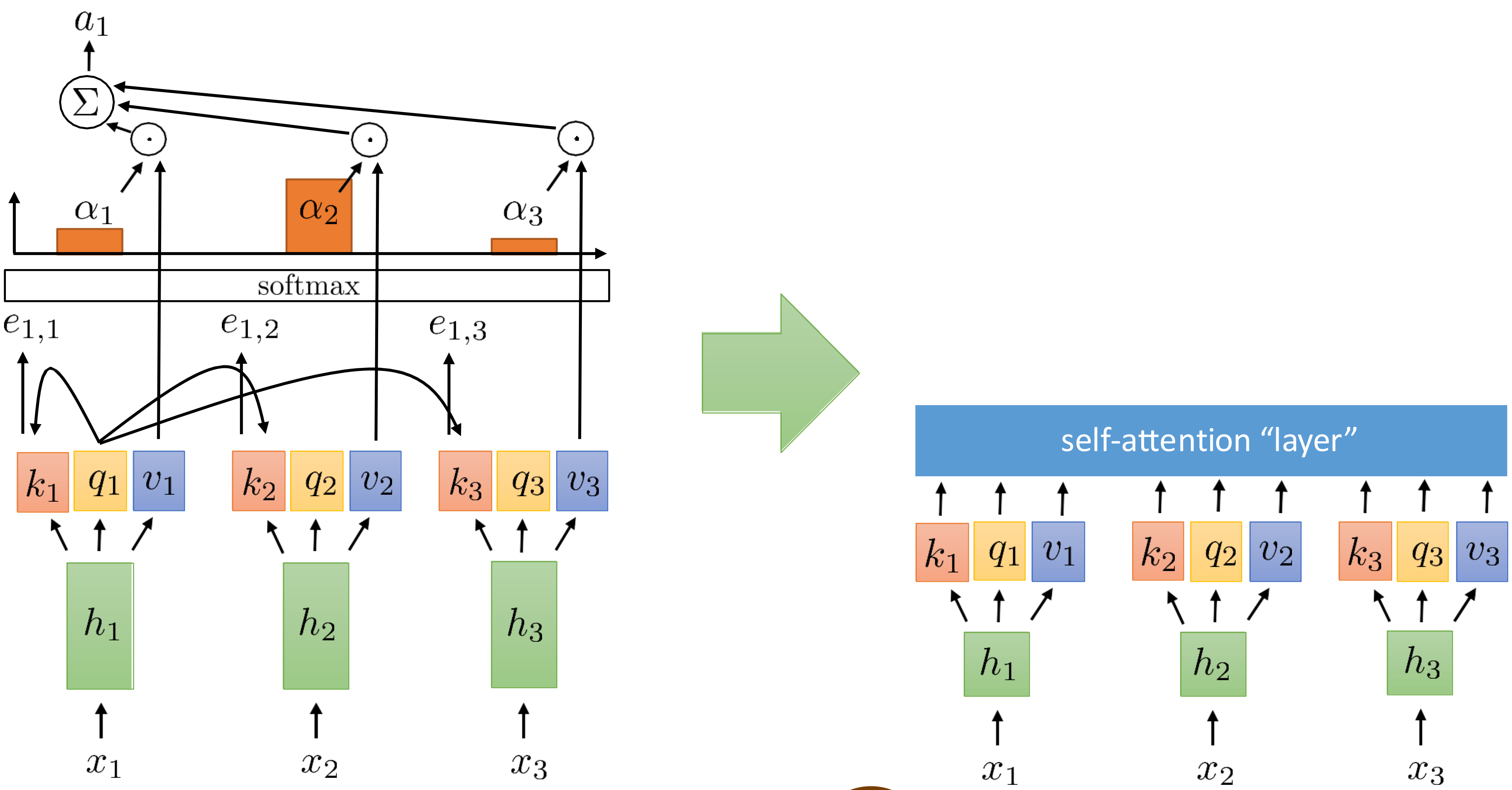

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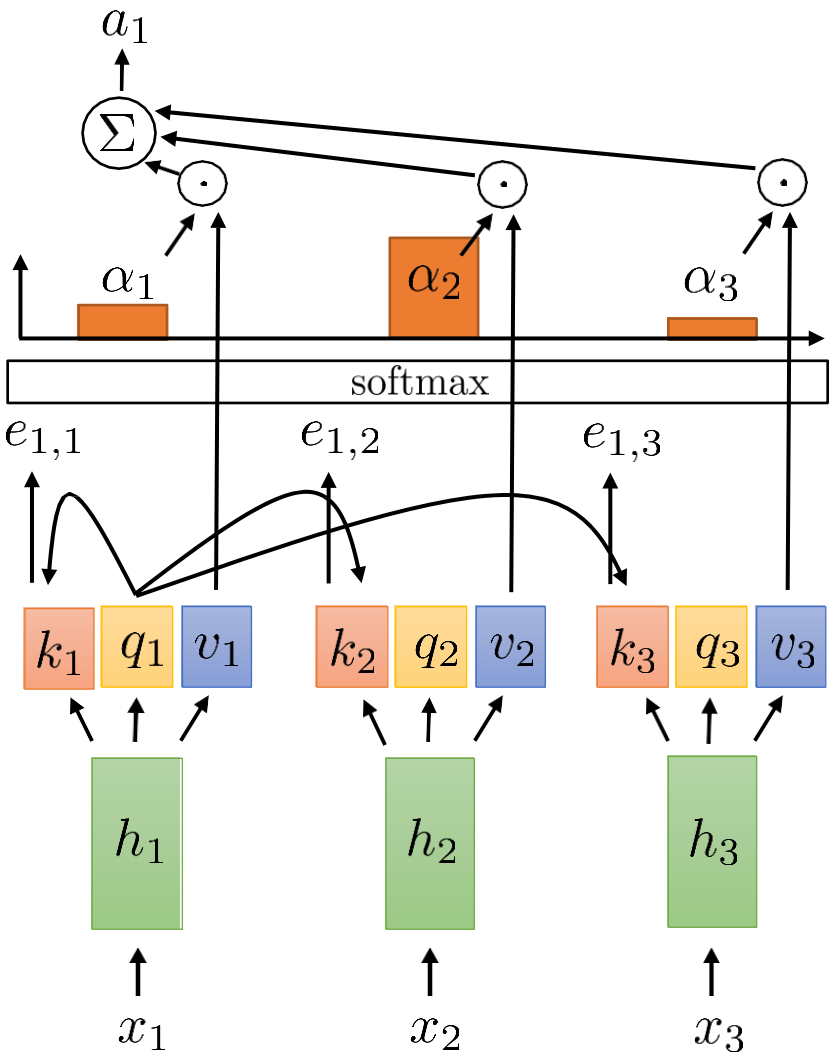

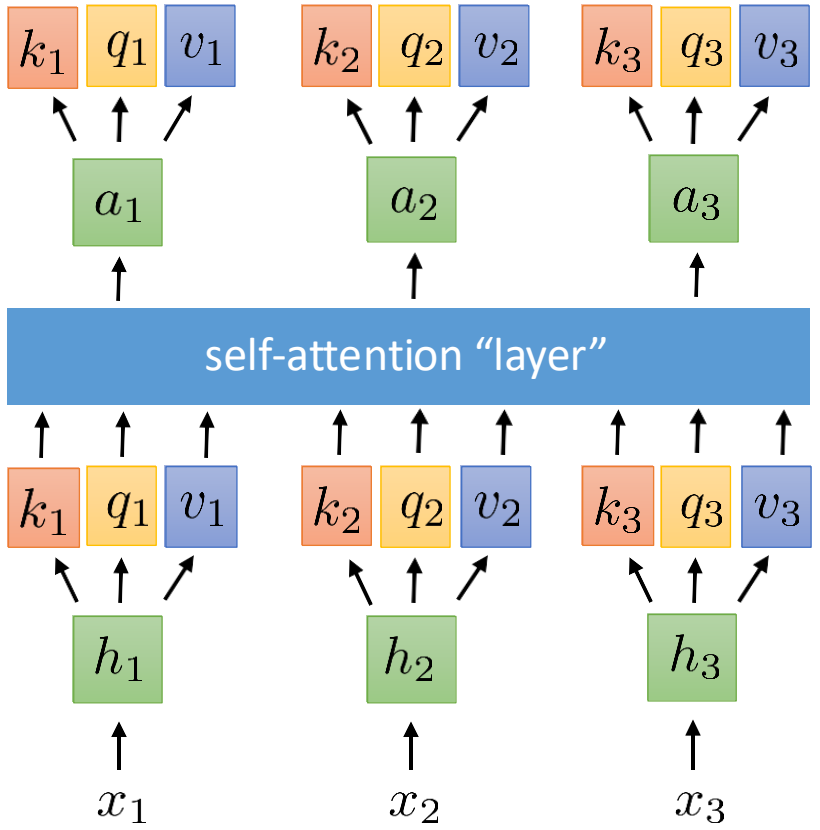

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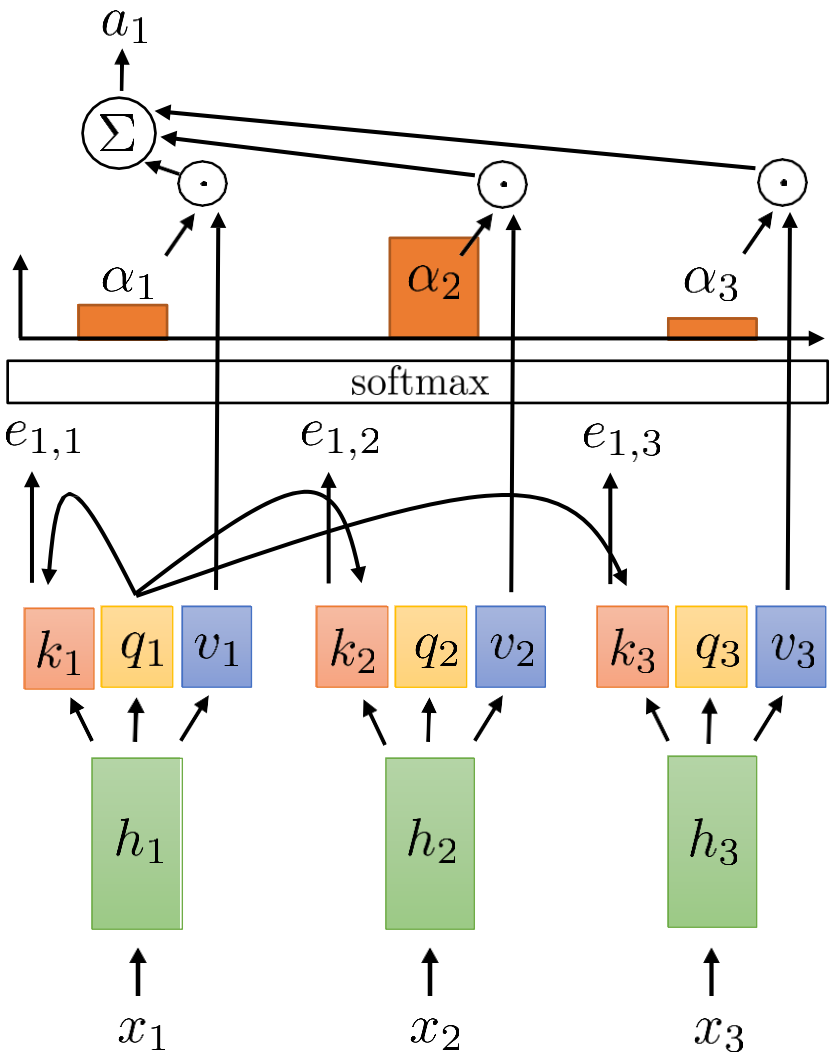

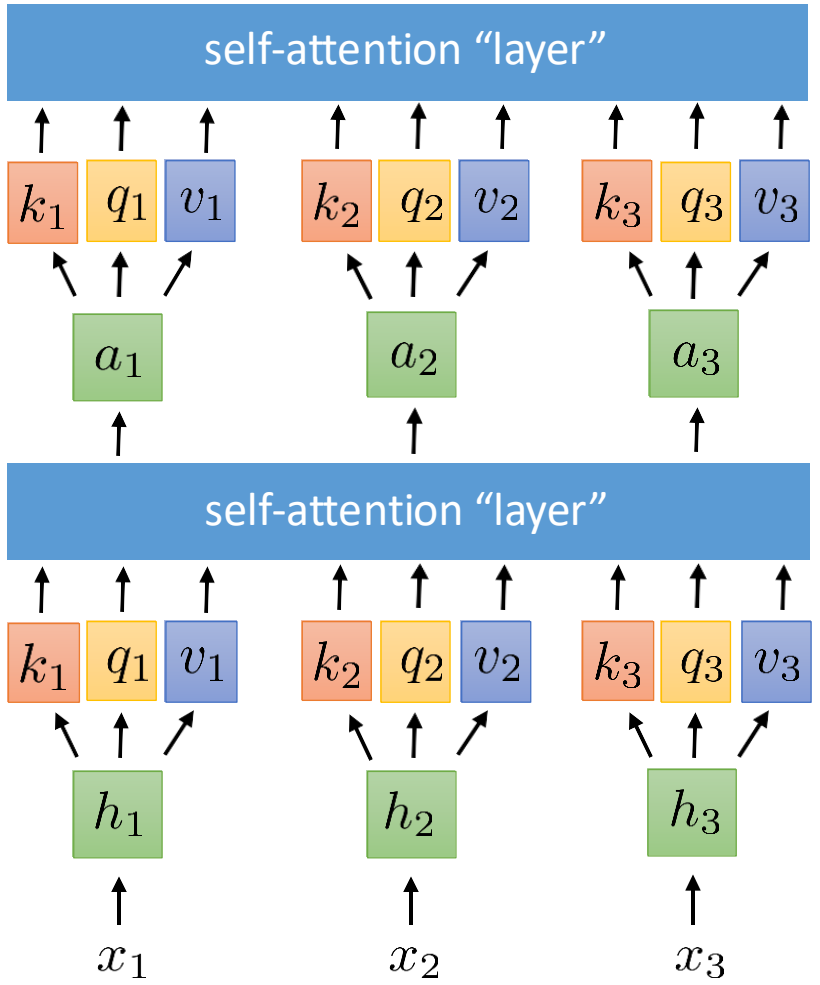

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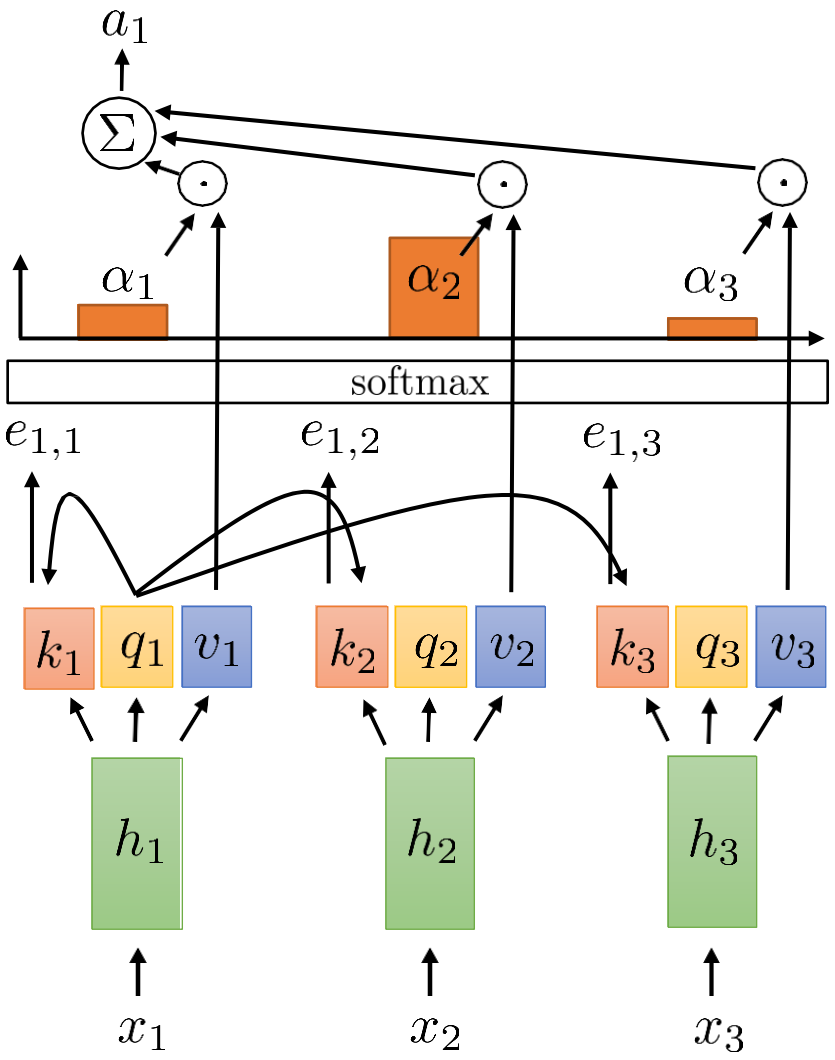

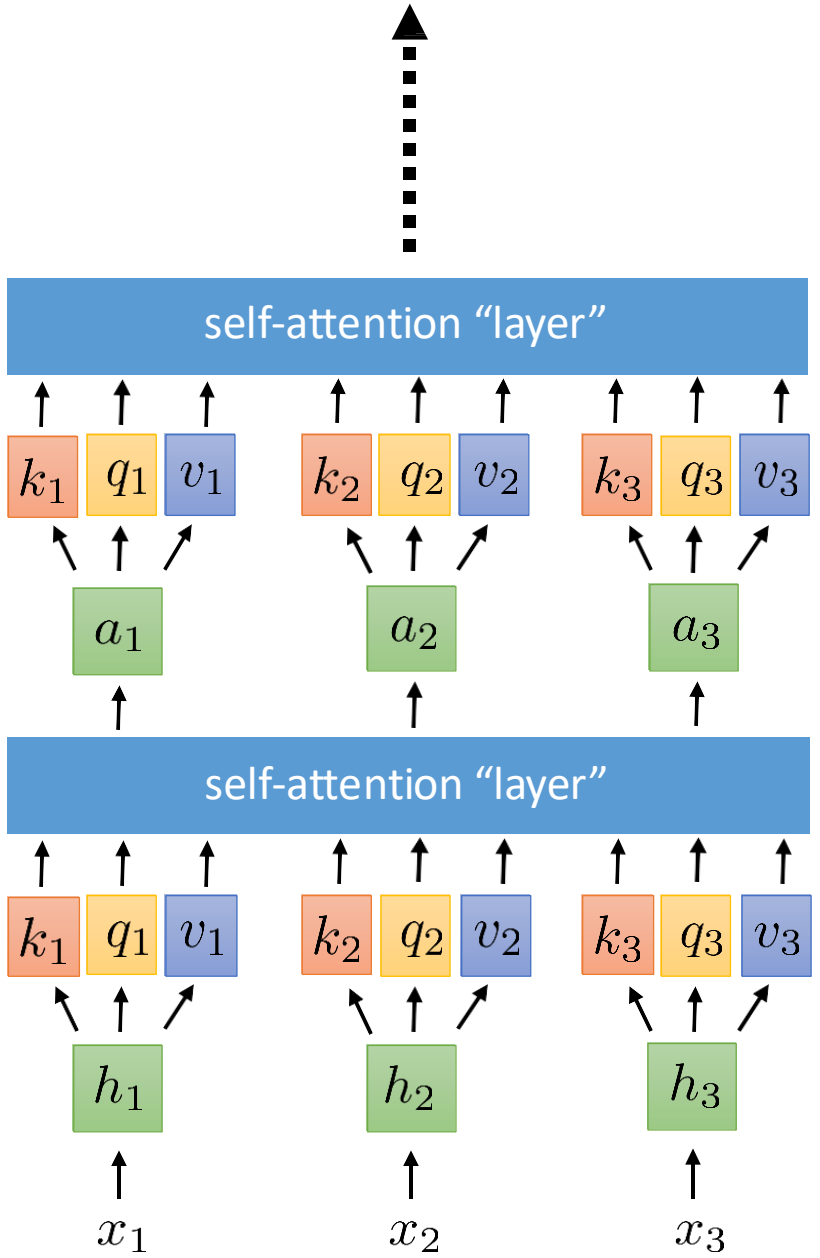

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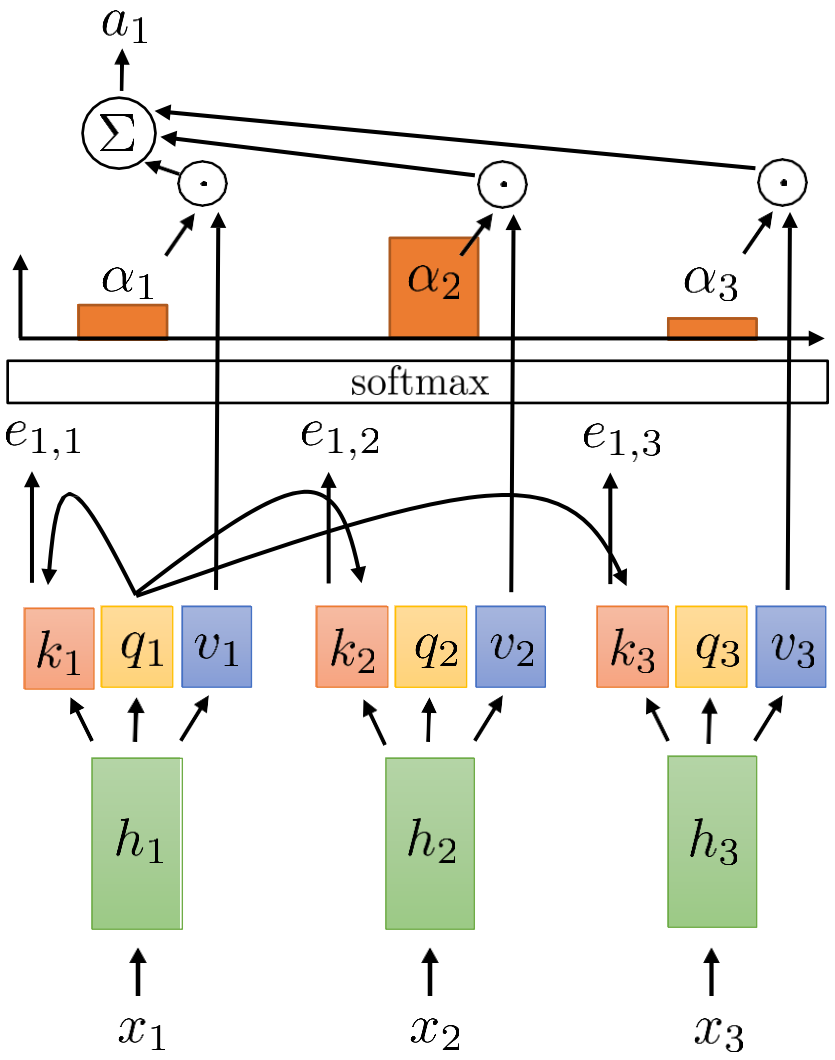

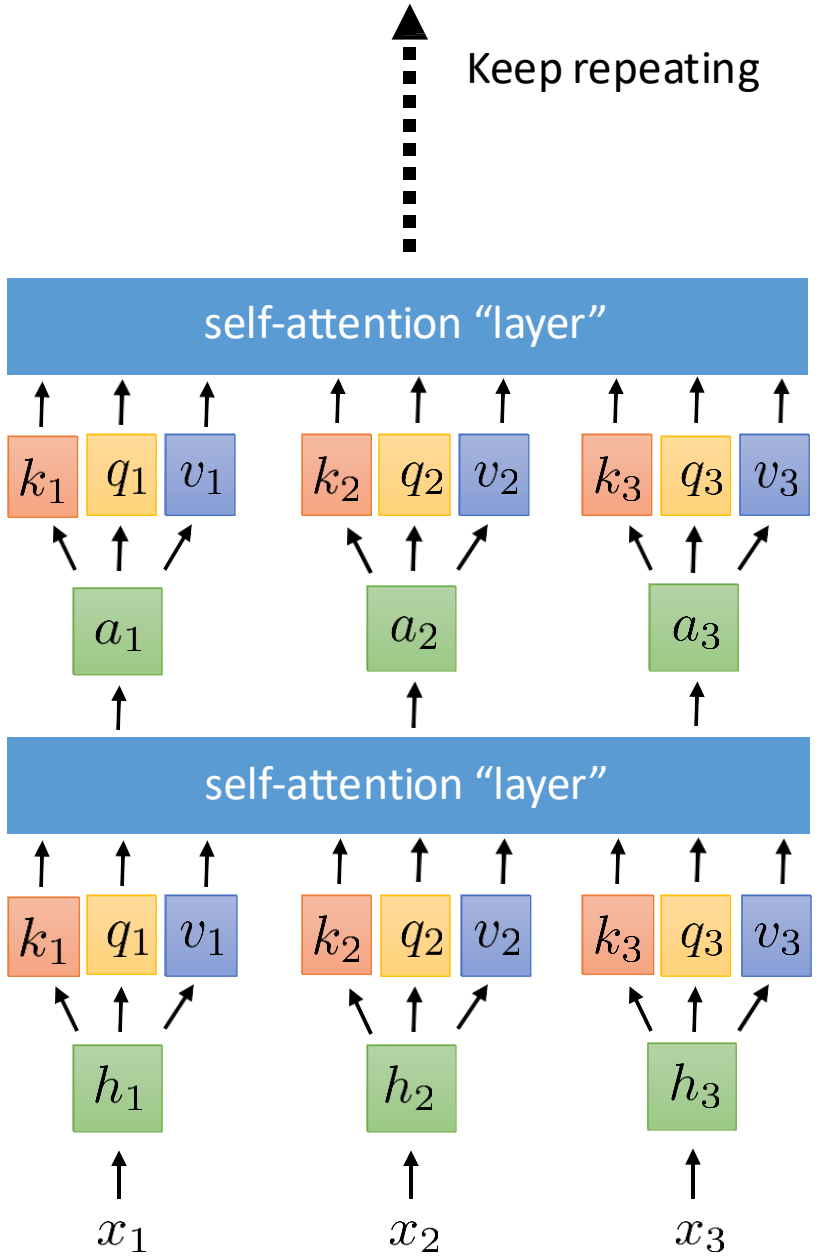

From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called **Transformers**.
- To address a few key limitations, we need to add certain elements:

1. Positional encoding	addresses lack of sequence information
2. Multi-headed attention	allows querying multiple positions at each layer
3. Adding nonlinearities	so far, each successive layer is *linear* in the previous one
4. Masked decoding	how to prevent attention lookups into the future?

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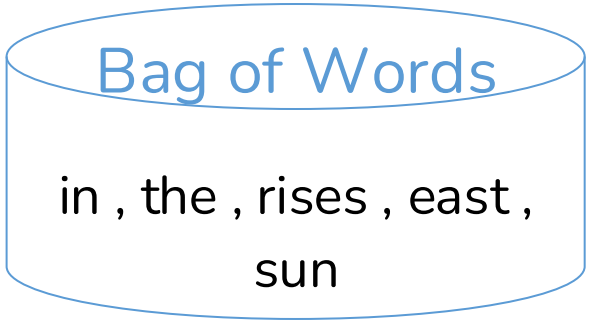
Positional Encoding - Motivation

- **Problem :** Self-attention processes all the elements of a sequence in parallel without any regard for their order.
  - Example : the sun rises in the east
  - Permuted version : rises in the sun the east

    the east rises in the sun



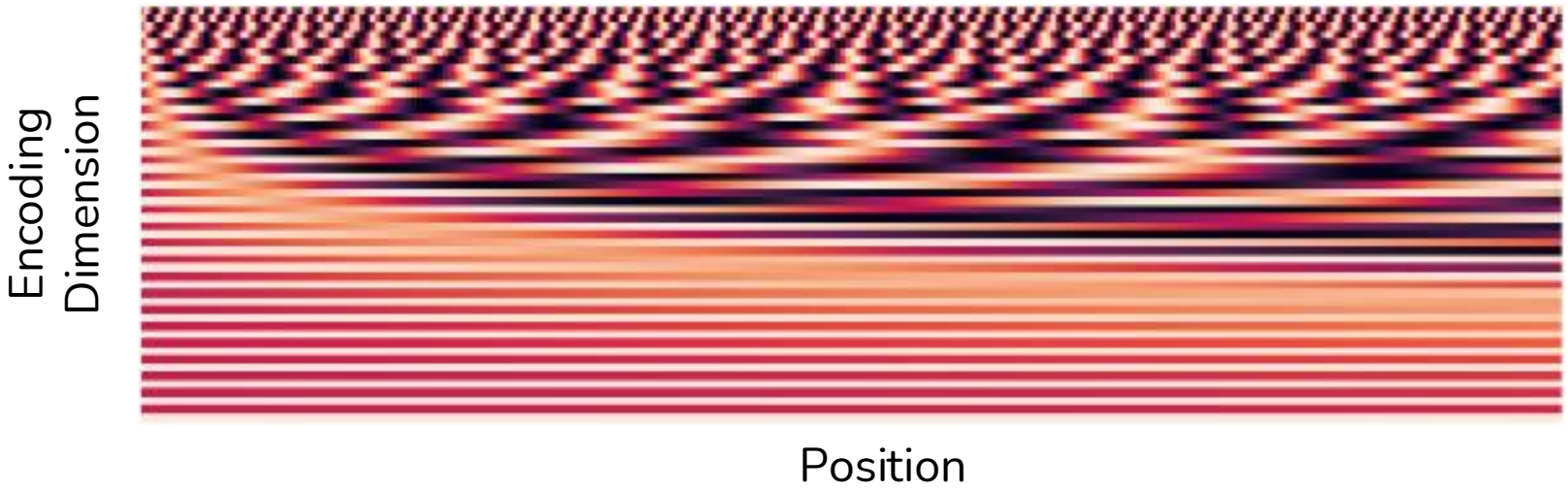
  - Self-attention is permutation invariant.
  - In natural language, it is important to take into account the order of words in a sentence.
- **Solution :** Explicitly add positional information to indicate where a word appears in a sequence

Sinusoidal Positional Encoding

- Helps it determine the position of each word (absolute positional information), or the distance between different words in the sequence(relative positional information)
- The frequency decreases along the encoding dimension.

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$



From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called transformers.
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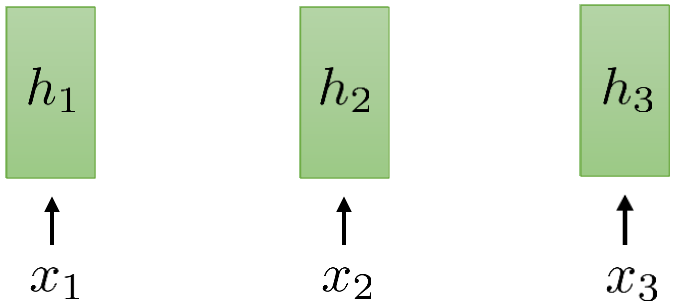
4. Masked decoding	how to prevent attention lookups into the future?

Multi-Head Attention

Given that we're fully depending on attention now, it could be beneficial to include more than one time step.



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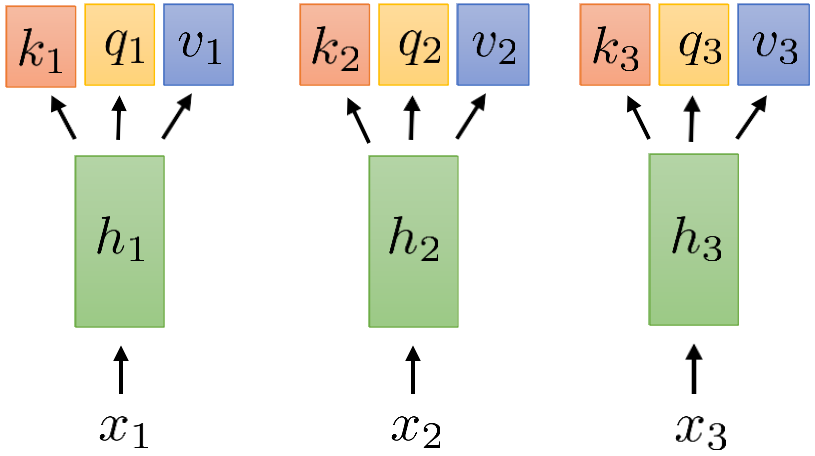

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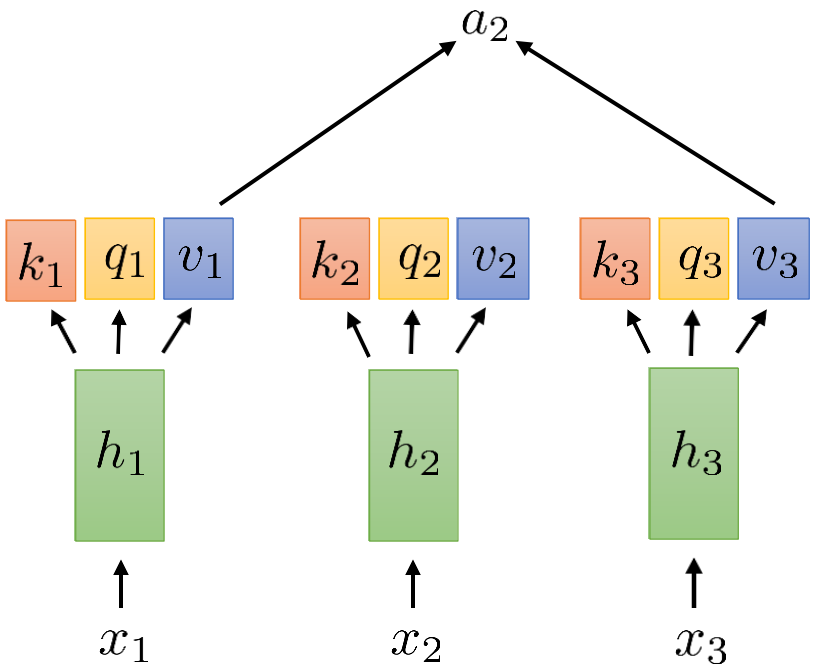

$$a_l = \sum_t \alpha_{l,t} v_t$$

Due to the softmax function, this will be heavily influenced by a single value.

Multi-Head Attention

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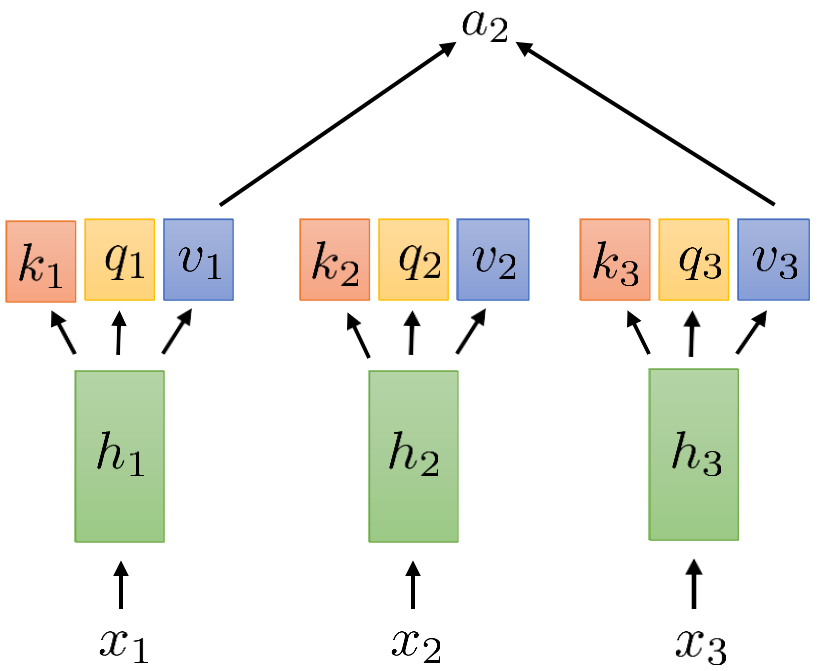


$$a_l = \sum_t \alpha_{l,t} v_t$$

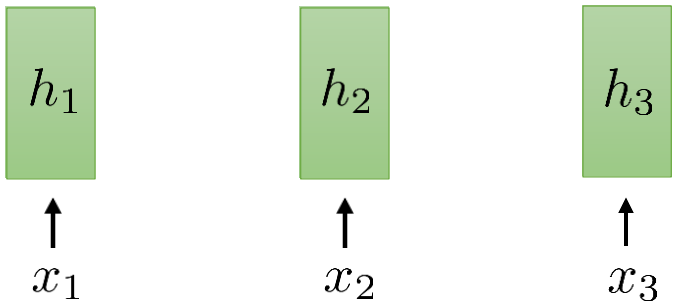
Due to the softmax function, this will be heavily influenced by a single value.

$$e_{l,t} = q_l \cdot k_t$$

It's challenging to clearly specify that you want two distinct elements, like the subject and object in a sentence.

Multi-Head Attention

Solution: Use multiple keys, queries, and values for each time step



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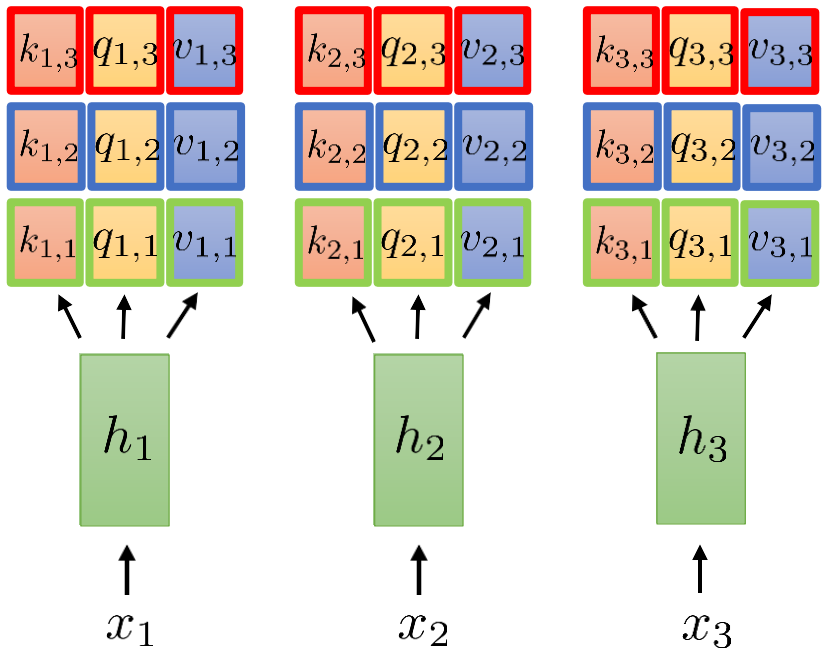

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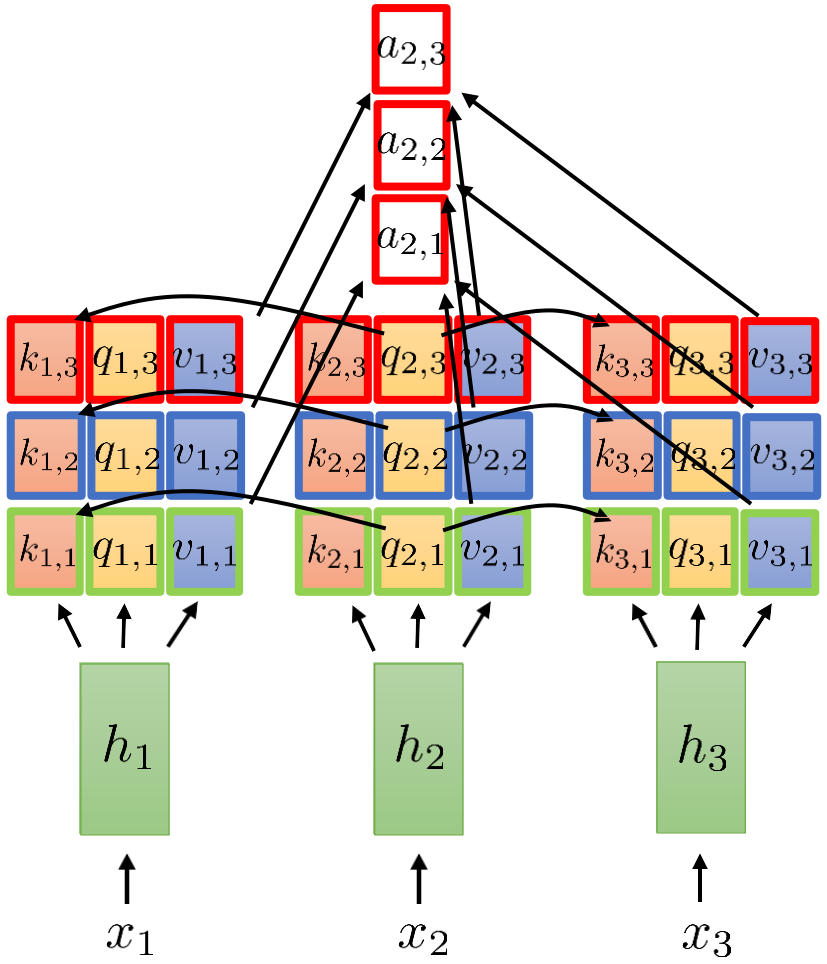

full attention vector formed by concatenation:

$$a_2 = \begin{bmatrix} a_{2,1} \\ a_{2,2} \\ a_{2,3} \end{bmatrix}$$

compute weights **independently** for each head

$$e_{l,t,i} = q_{l,i} \cdot k_{l,i}$$

$$\alpha_{l,t,i} = \exp(e_{l,t,i}) / \sum_{t'} \exp(e_{l,t',i})$$

$$a_{l,i} = \sum_{t} \alpha_{l,t,i} v_{t,i}$$

From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called transformers.
- To address a few key limitations, we need to add certain elements:

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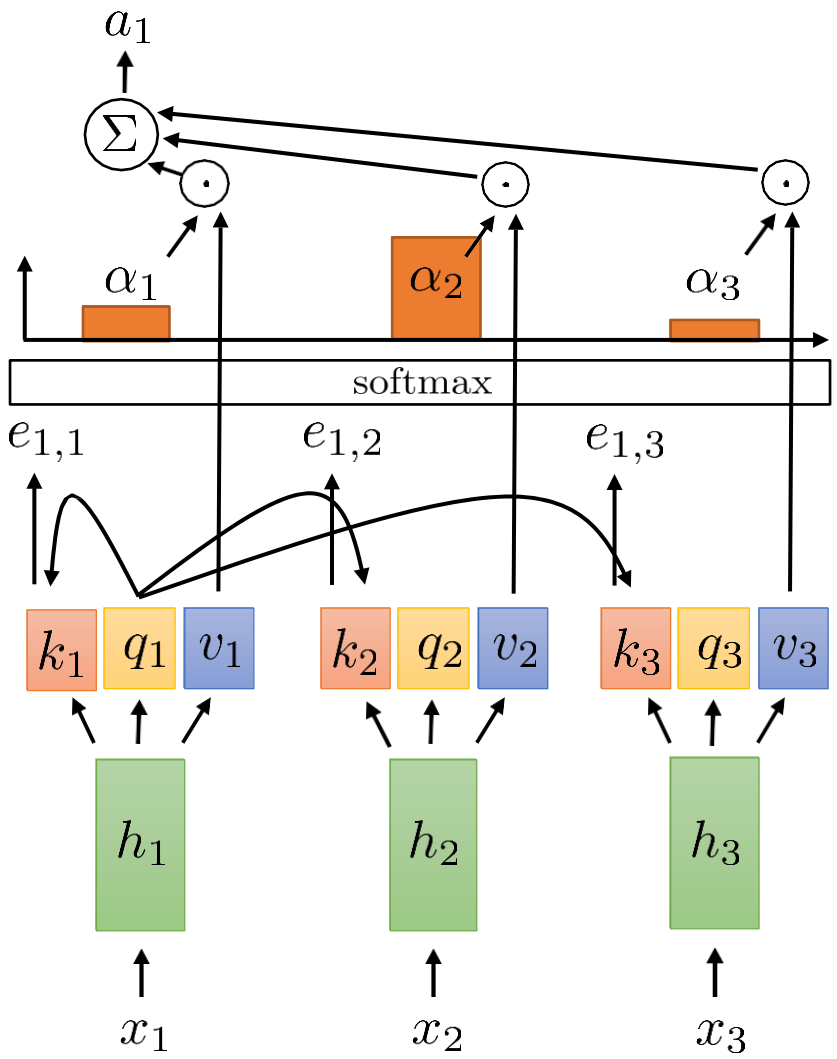
4. Masked decoding	how to prevent attention lookups into the future?

Self-Attention Is “Linear”



$$k_t = W_k h_t \qquad q_t = W_q h_t \qquad v_t = W_v h_t$$

$$\alpha_{l,t} = \exp(e_{l,t}) / \sum_{t'} \exp(e_{l,t'})$$

$$e_{l,t} = q_l \cdot k_t$$

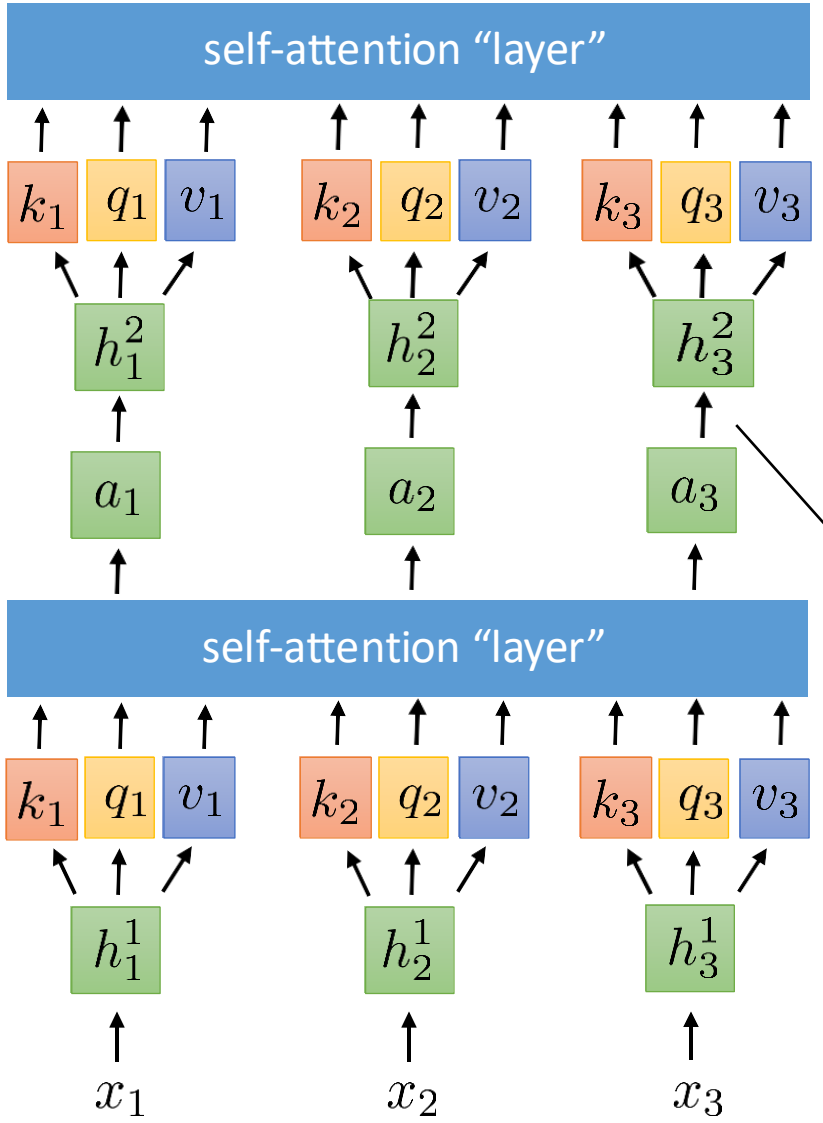
$$a_l = \sum_t \alpha_{l,t} v_t = \sum_t \alpha_{l,t} W_v h_t = W_v \sum_t \alpha_{l,t} h_t$$

linear transformation

non-linear weights

Problem: Every self-attention layer is a linear transformation of the previous layer with non-linear weights.

Position-wise Feed-Forward Networks



- **Solution :** Make the model more expressive is by alternating use of self-attention and non-linearity.
- Non-linearity is incorporated by means of a feed-forward network which consists of two linear transformations with a ReLU activation in between.

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

- The same non-linearity is utilized across various positions but they differ from layer to layer.

From Self-Attention to Transformers

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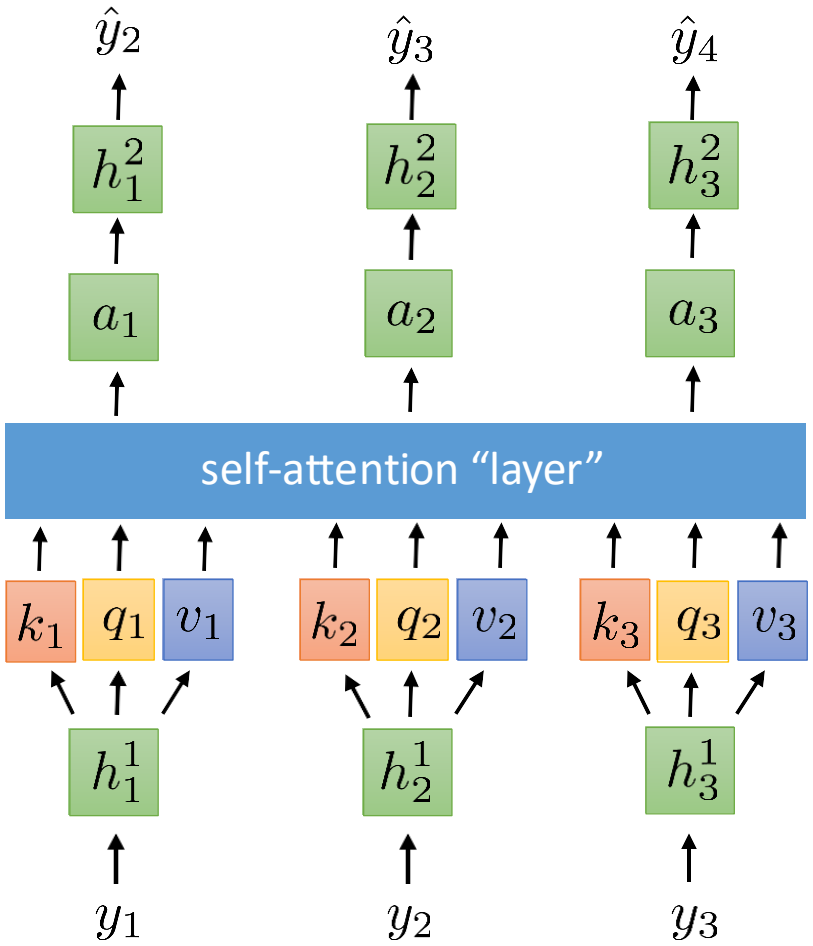
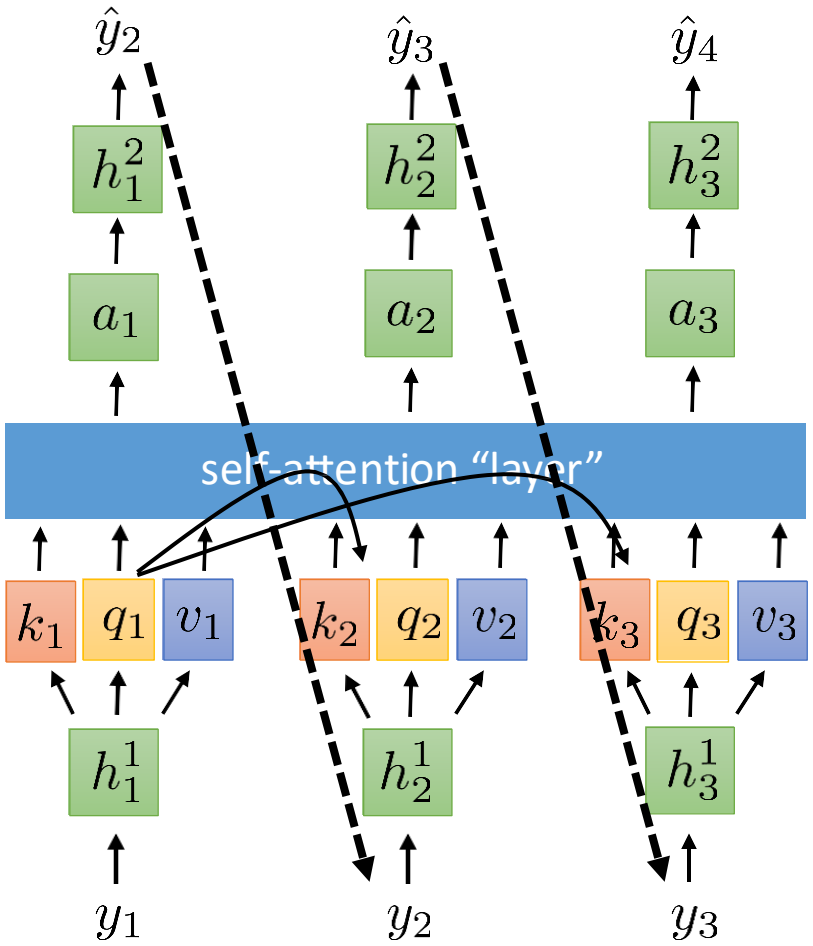
Self-attention can see the future!



A **crude** self-attention “language model”:

In practice, there would be several alternating self-attention layers and position-wise feedforward networks

Self-attention can see the future!



A **crude** self-attention "language model":

In practice, there would be several alternating self-attention layers and position-wise feedforward networks

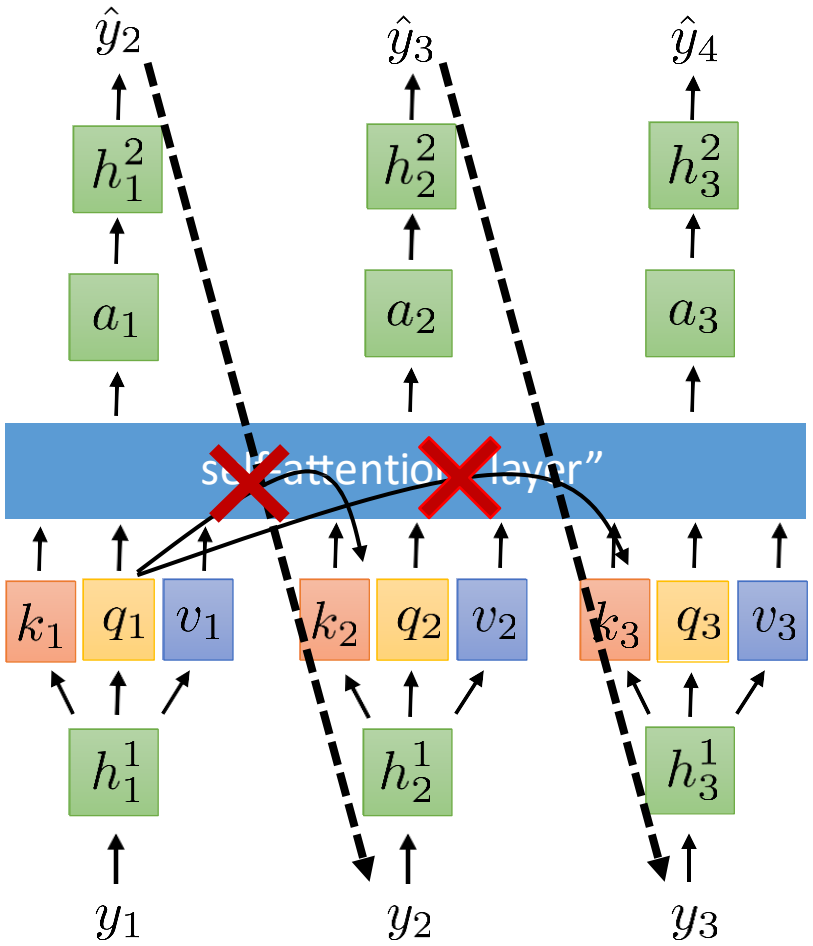
Big problem: self-attention at step 1 can look at the value at steps 2 & 3, which is based on the **inputs** at steps 2 & 3

At test time (when decoding), the **inputs** at steps 2 & 3 will be based on the output at step 1…

…which requires knowing the **input** at steps 2 & 3

Masked Attention

A **crude** self-attention "language model":



At test time (when decoding), the **inputs** at steps 2 & 3 will be based on the output at step 1…

…which requires knowing the **input** at steps 2 & 3

Must allow self-attention into the **past**…

…but not into the **future**

Easy solution:

~~$e_{l,t} = q_l \cdot k_t$~~

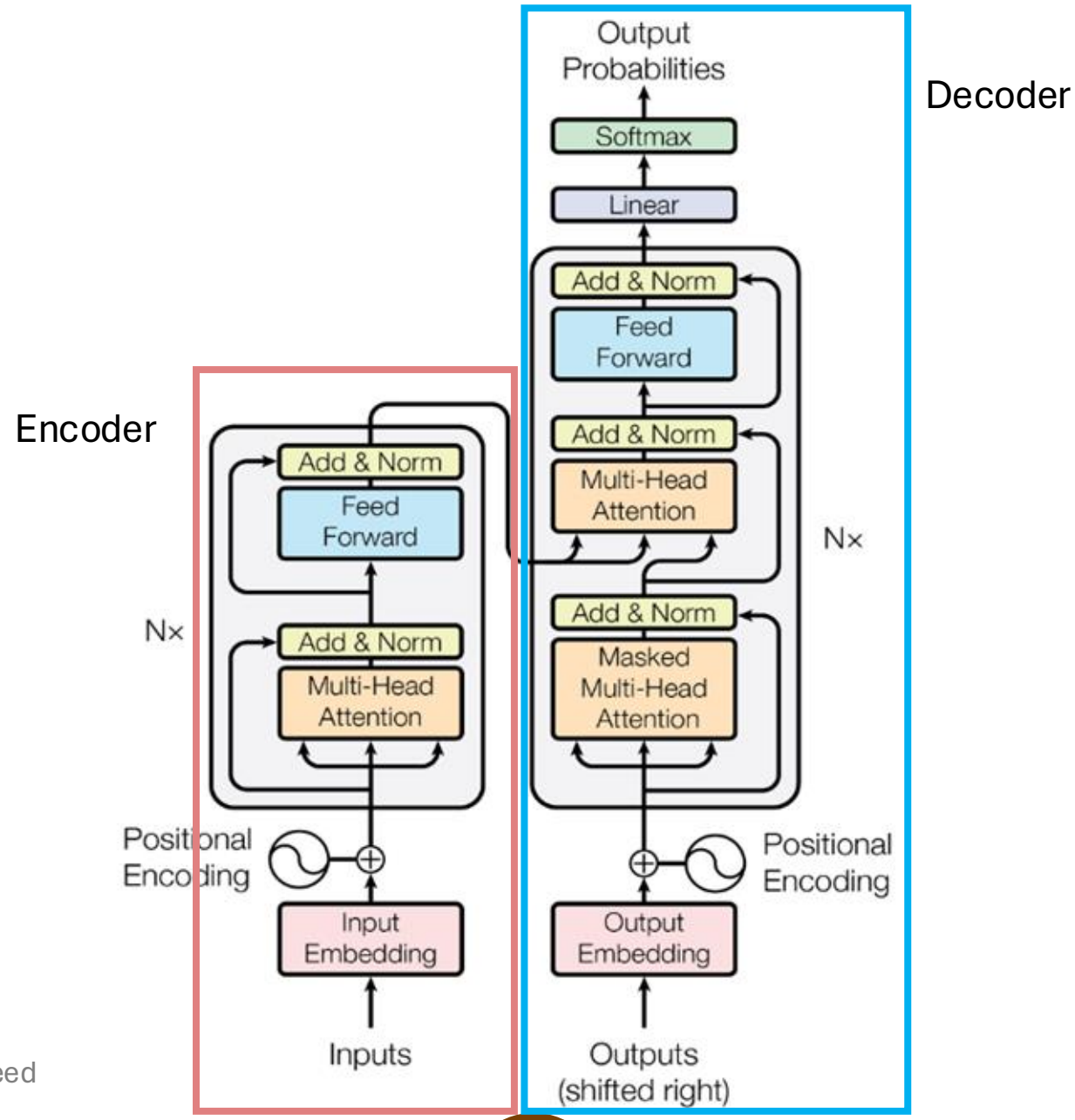
$$e_{l,t} = \begin{cases} q_l \cdot k_t & \text{if } l \geq t \\ -\infty & \text{otherwise} \end{cases}$$

in practice:

just replace $\exp(e_{l,t})$ with 0 if $l < t$

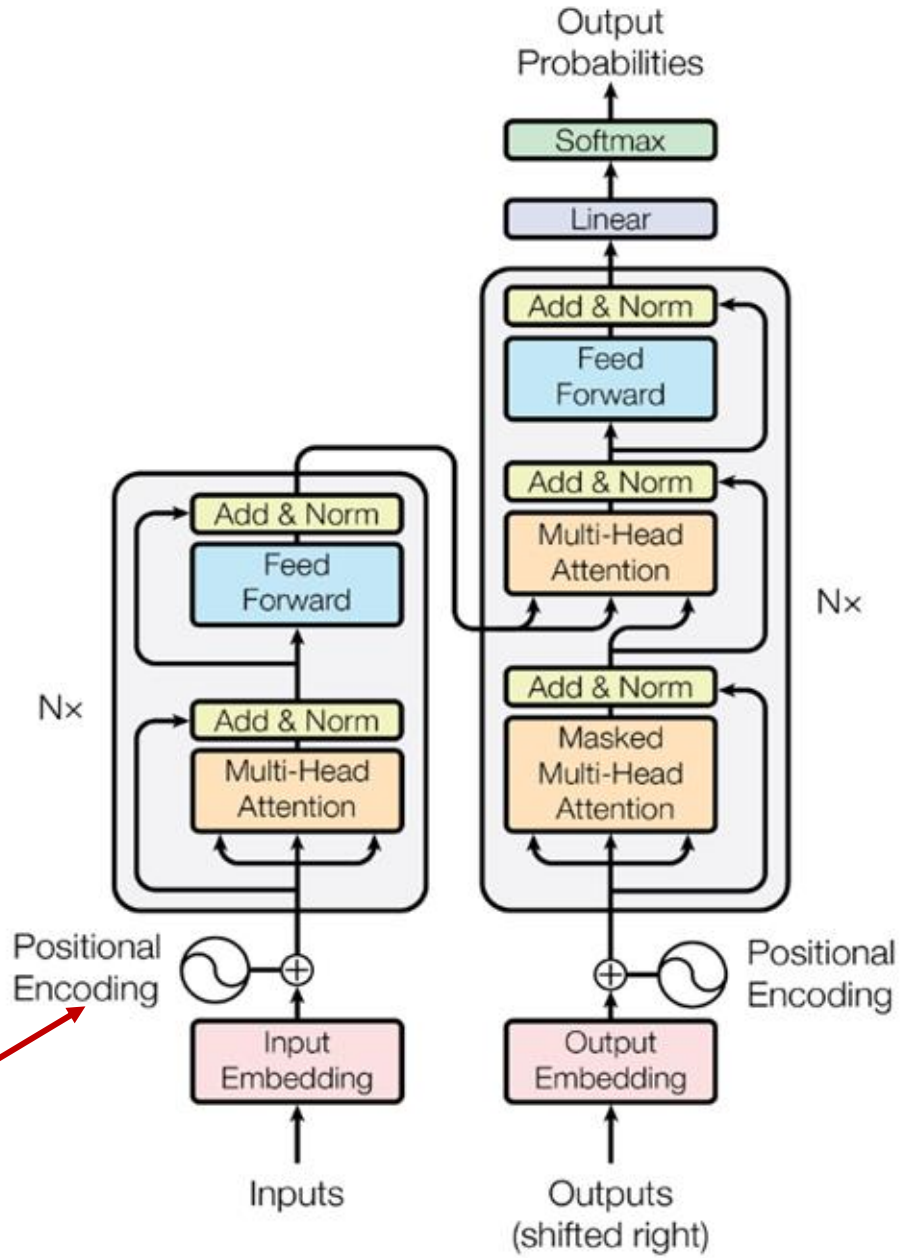
inside the softmax

Transformer Architecture



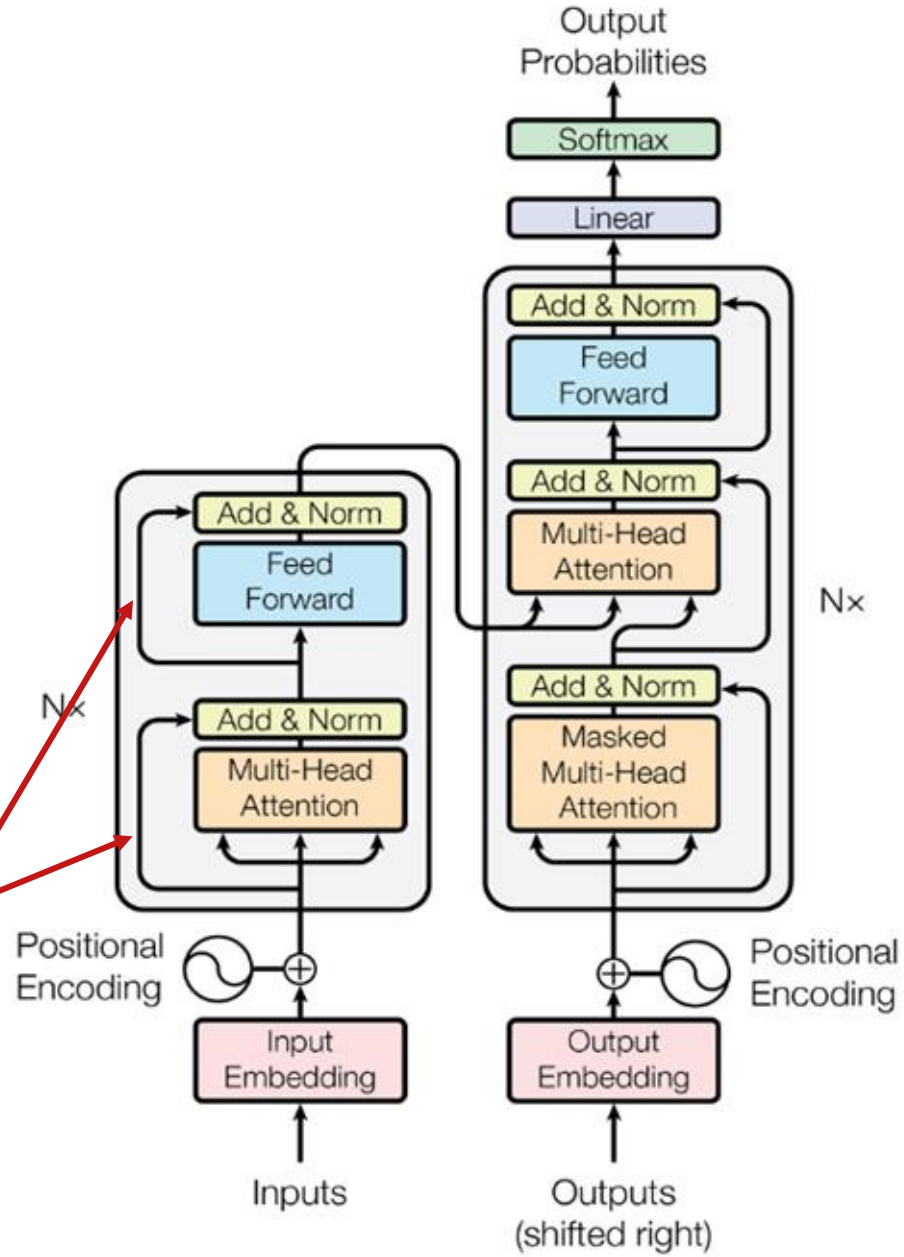
Source of Image : Attention is all you need (Vaswani t al., 2017)

Transformer Architecture



Position embeddings are *added* to each word embedding. Otherwise, since we have no recurrence, our model is unaware of the position of a word in the sequence!

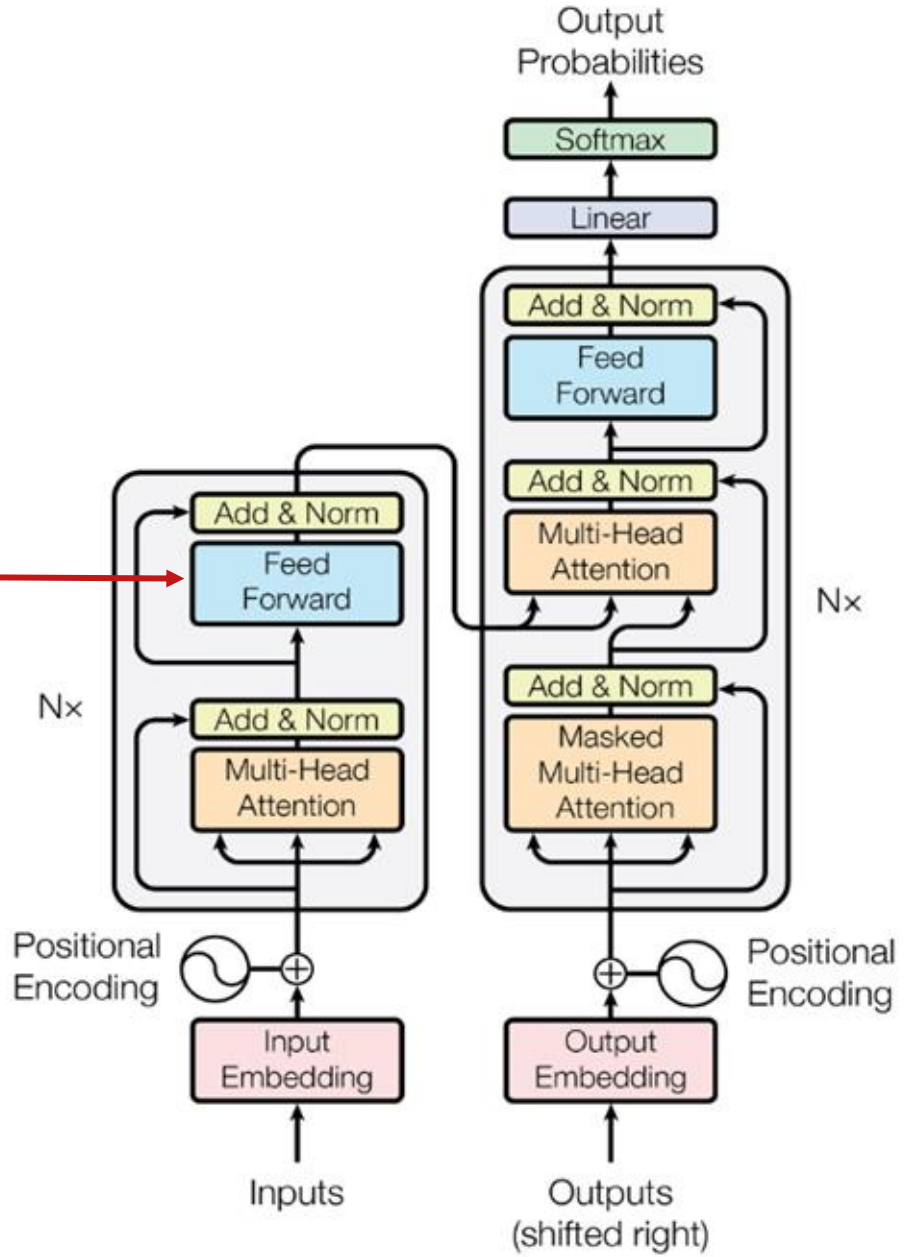
Transformer Architecture



Residual connections, which mean that we add the input to a particular block to its output, help improve gradient flow

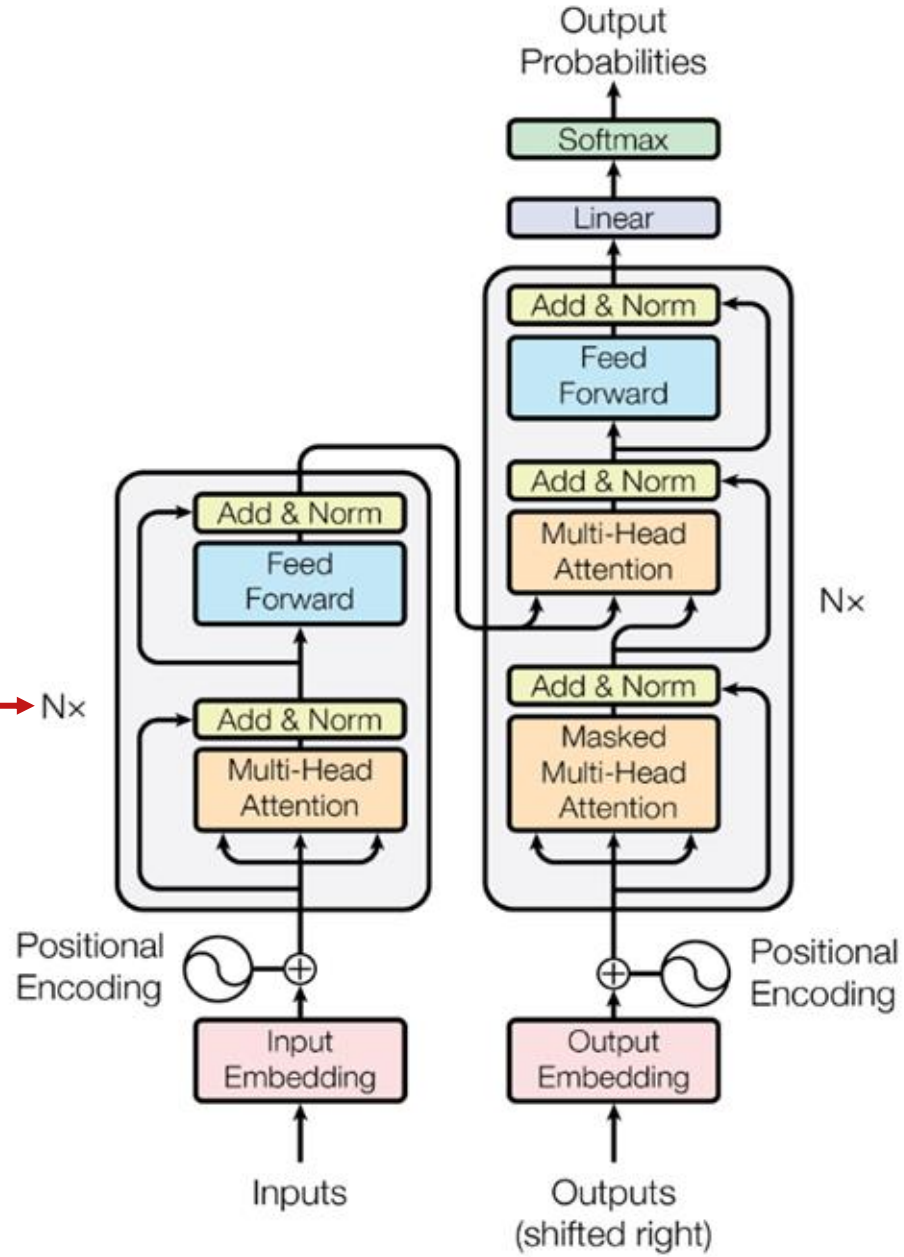
Transformer Architecture

A **feed-forward layer** on top of the attention- weighted averaged value vectors allows us to add more parameters / nonlinearity

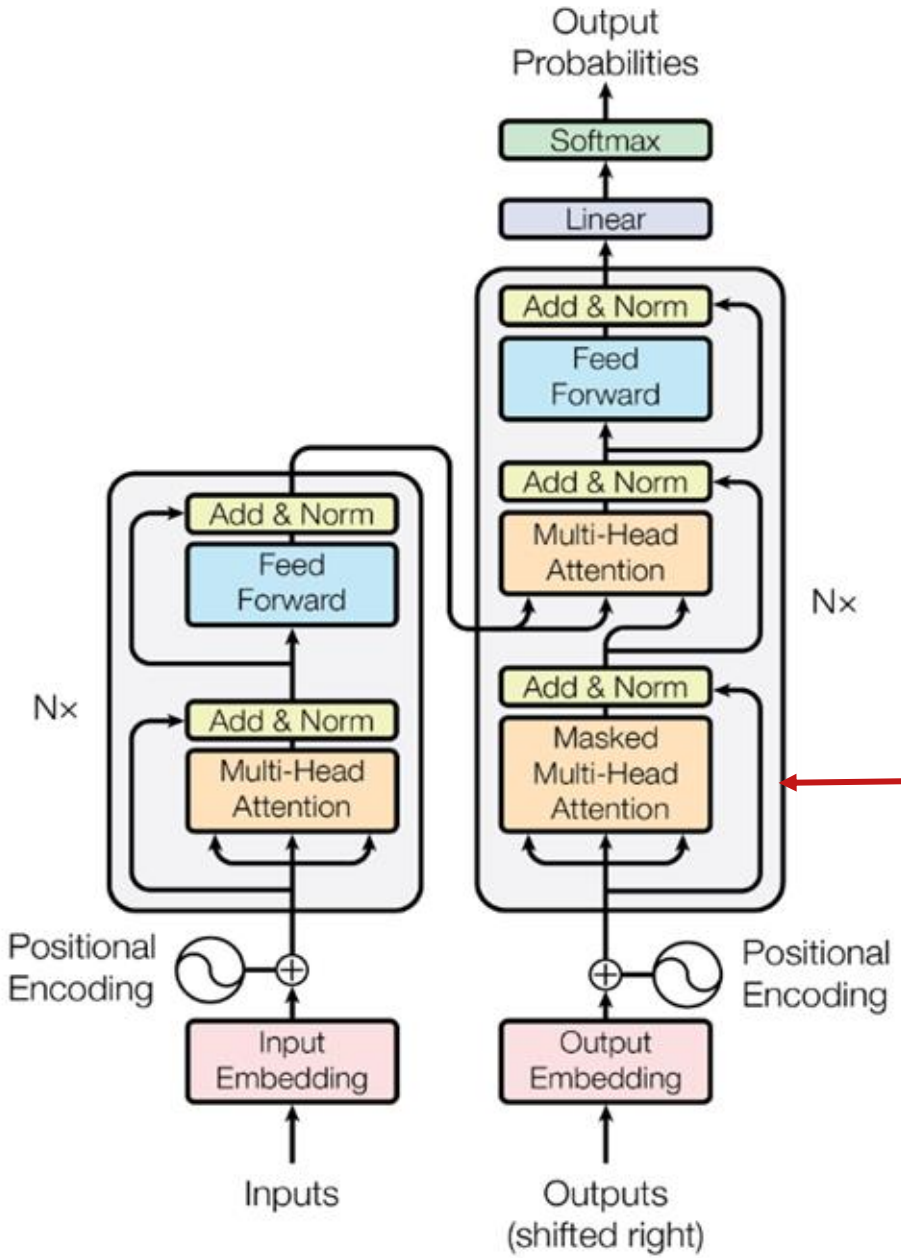


Transformer Architecture

We stack as many of these ***Transformer* blocks** on top of each other as we can (bigger models are generally better given enough data!)

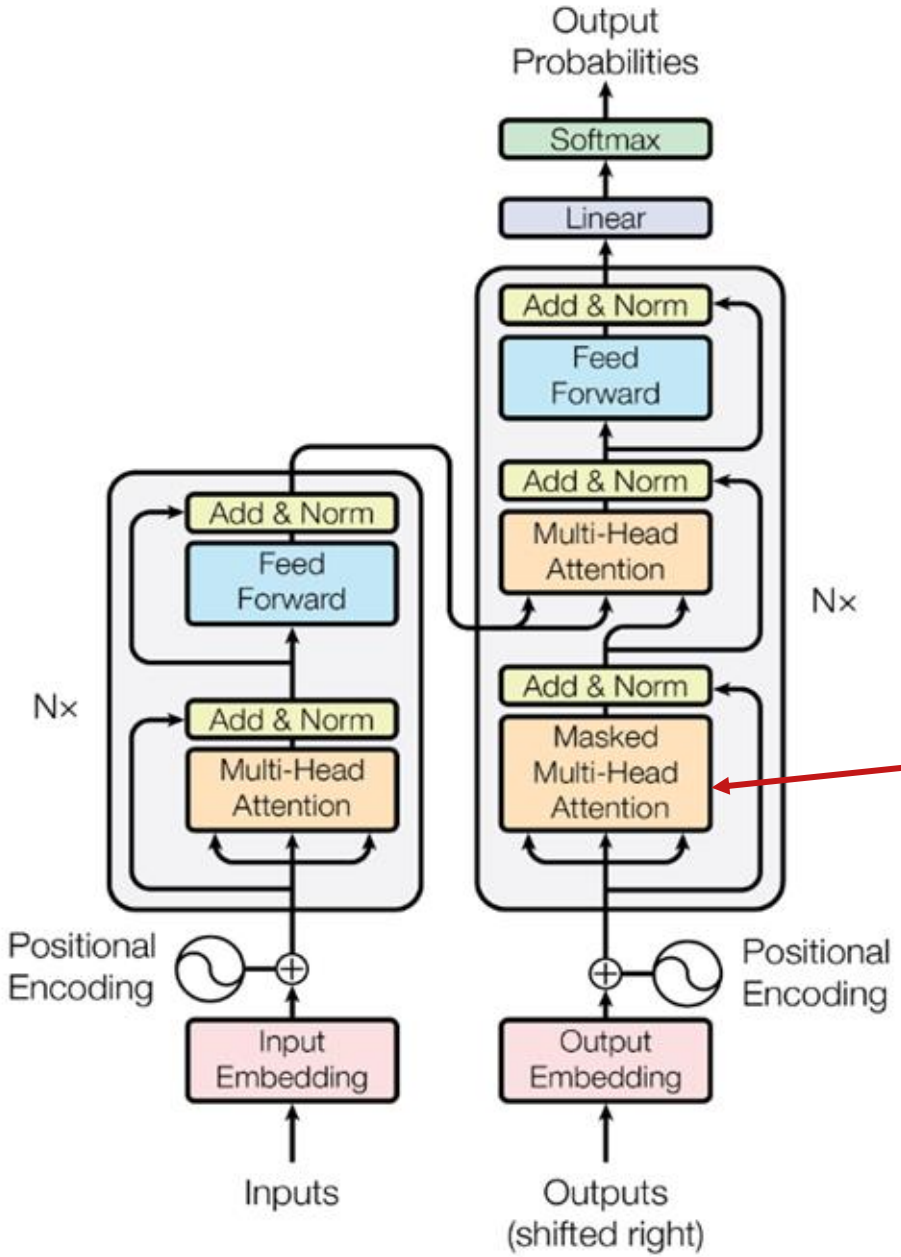


Transformer Architecture



Moving onto the decoder, which takes in English sequences that have been shifted to the right (e.g., *<START> schools opened their*)

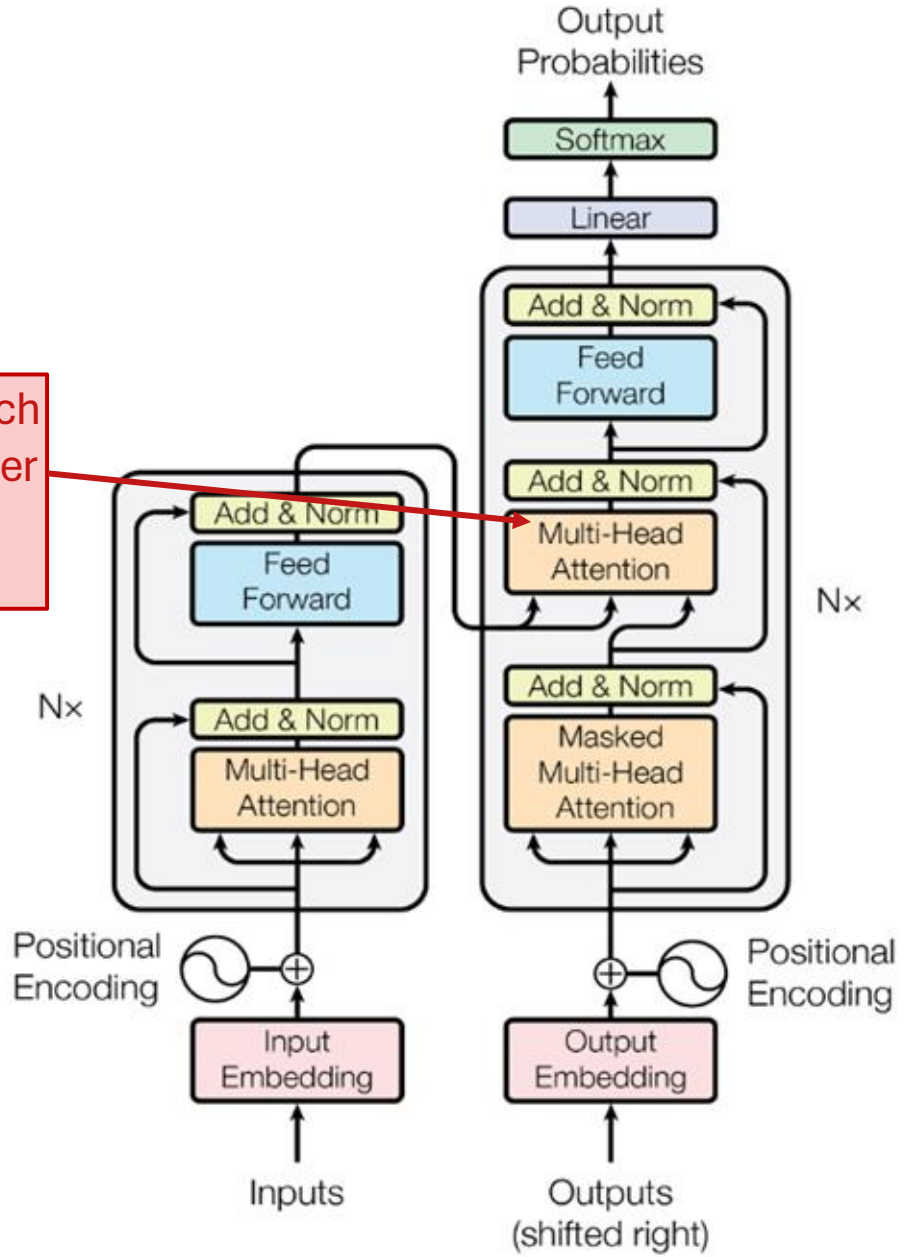
Transformer Architecture



We first have an instance of **masked self attention**. Since the decoder is responsible for predicting the English words, we need to apply masking as we saw before.

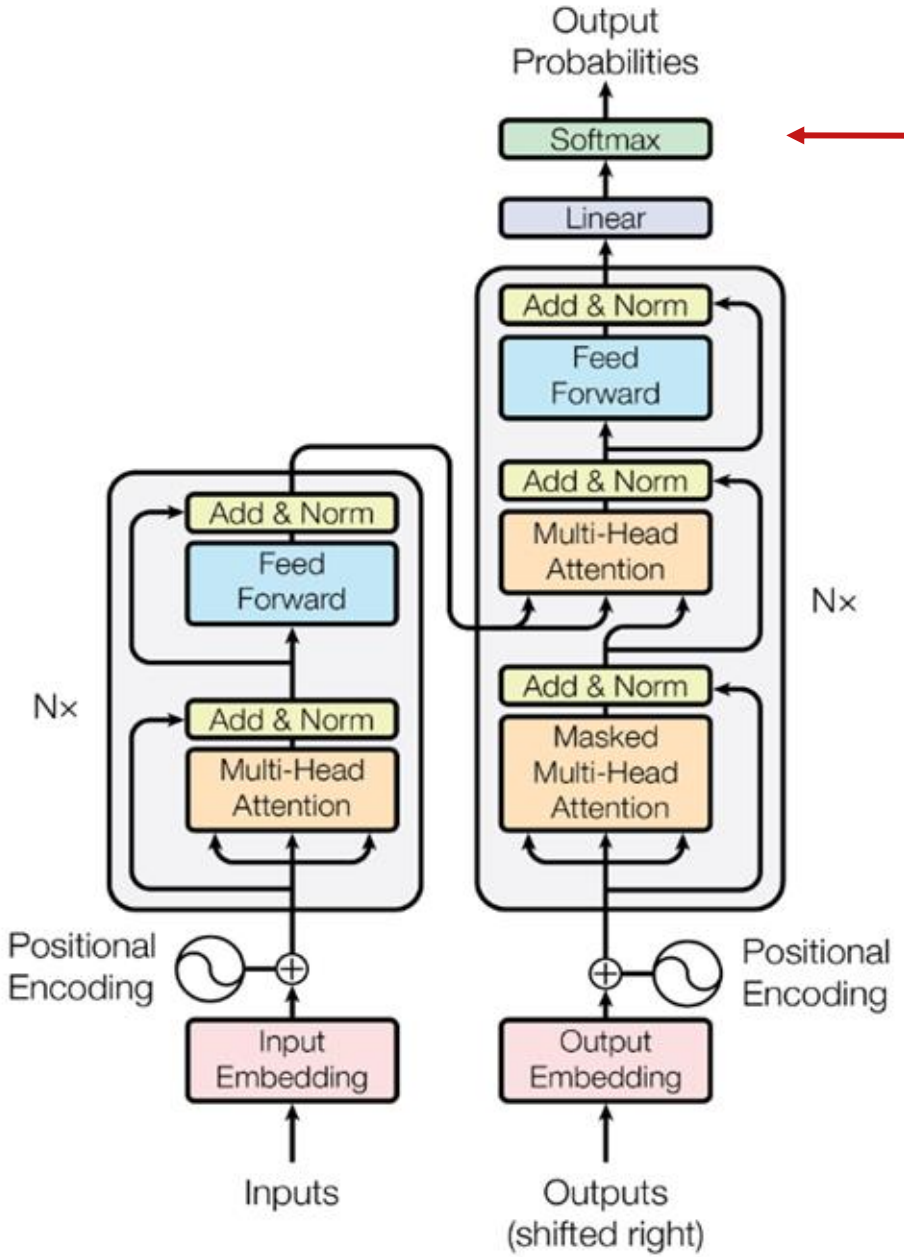
Transformer Architecture

Now, we have ***cross attention***, which connects the decoder to the encoder by enabling it to attend over the encoder's final hidden states.



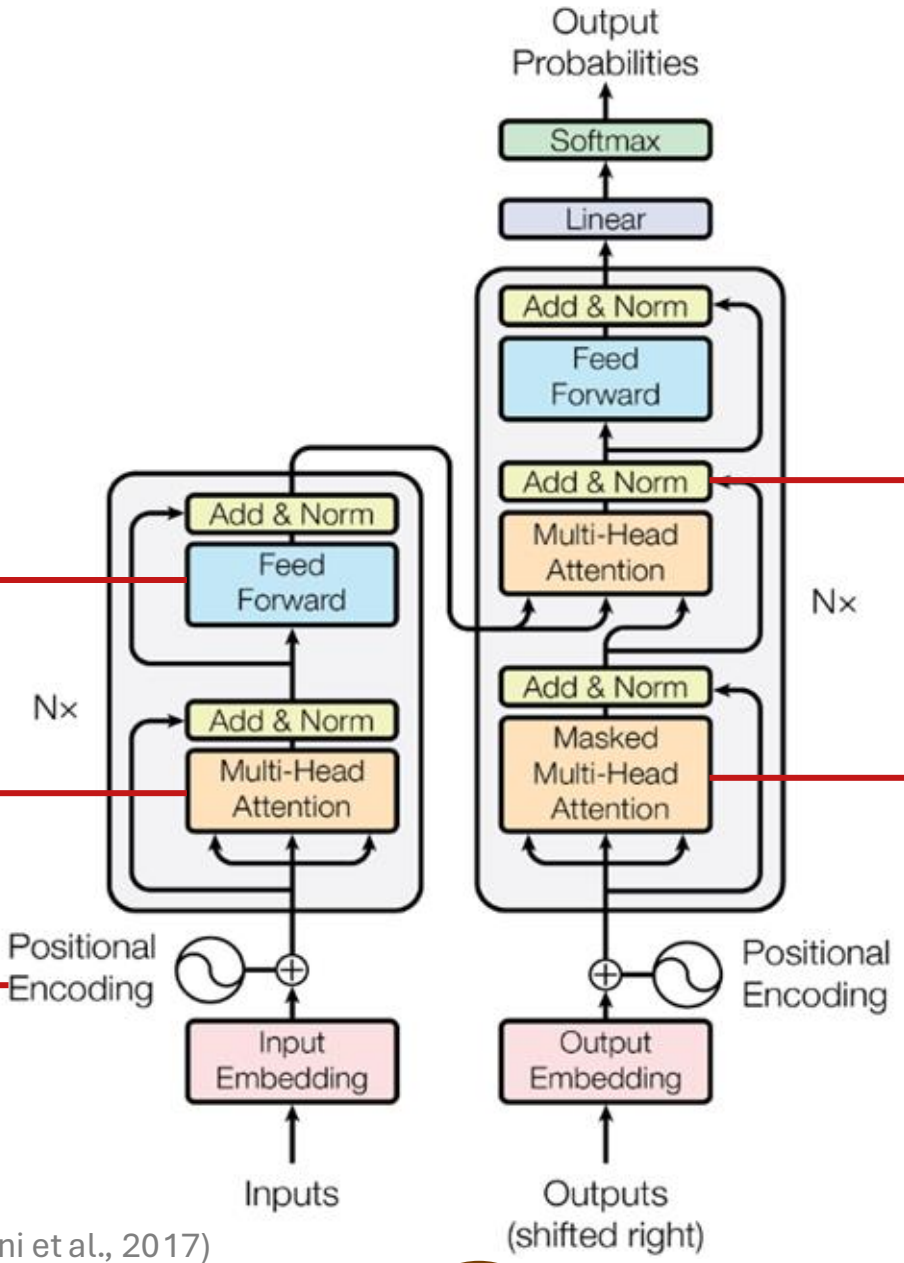
Source of Image : Attention is all you need (Vaswani et al., 2017)

Transformer Architecture



After stacking a bunch of these decoder blocks, we finally have our familiar **softmax** layer to predict the next English word.

Transformer Architecture



Adding non-linearities

Allows querying multiple positions at each layer

Adds positional information

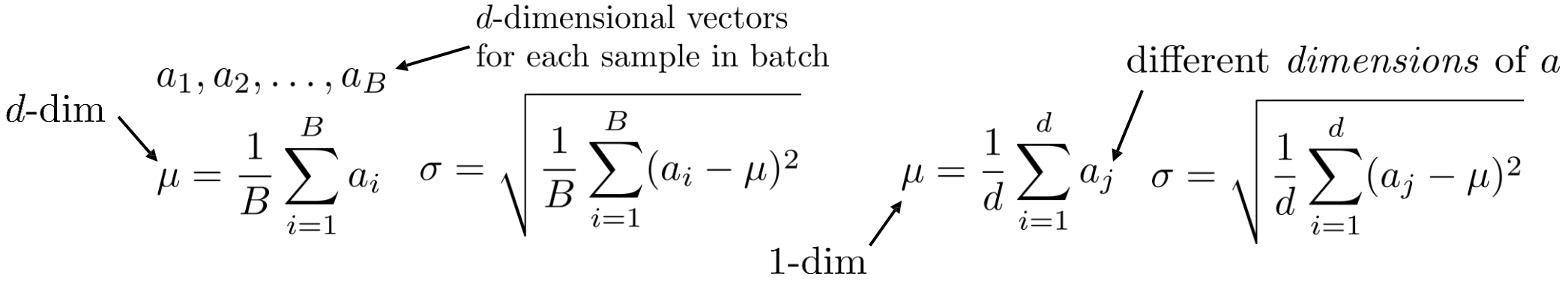
Reduces covariance shift and makes the system stable

Prevents attention lookups into the future while decoding

Source of Image : Attention is all you need (Vaswani et al., 2017)

Layer normalization

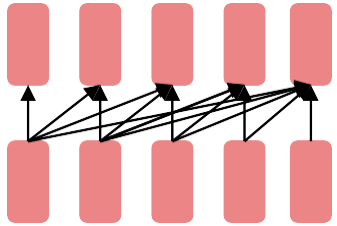
- **Main idea:** Batch normalization is quite beneficial, but it's challenging to apply with sequence models. The varying lengths of sequences make it difficult to normalize across a batch. Sequences can be very long, which often results in smaller batch sizes.
- **Solution:** Layer normalization



Pre-training Strategies

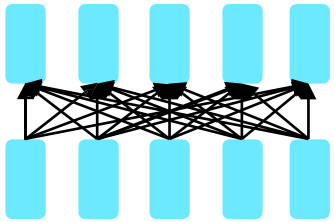
Pretraining for Three Types of Architectures

The neural architecture influences the type of pretraining, and natural use cases.



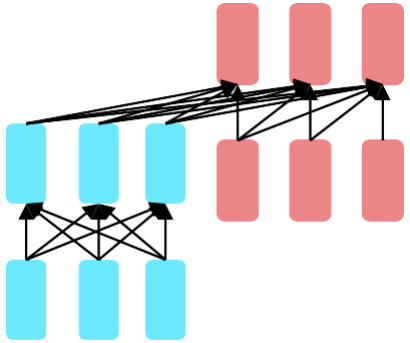
Decoders

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words



Encoders

- Gets bidirectional context – can condition on future!
- How do we pretrain them?



Encoder-Decoders

- Good parts of decoders and encoders?
- What's the best way to pretrain them?

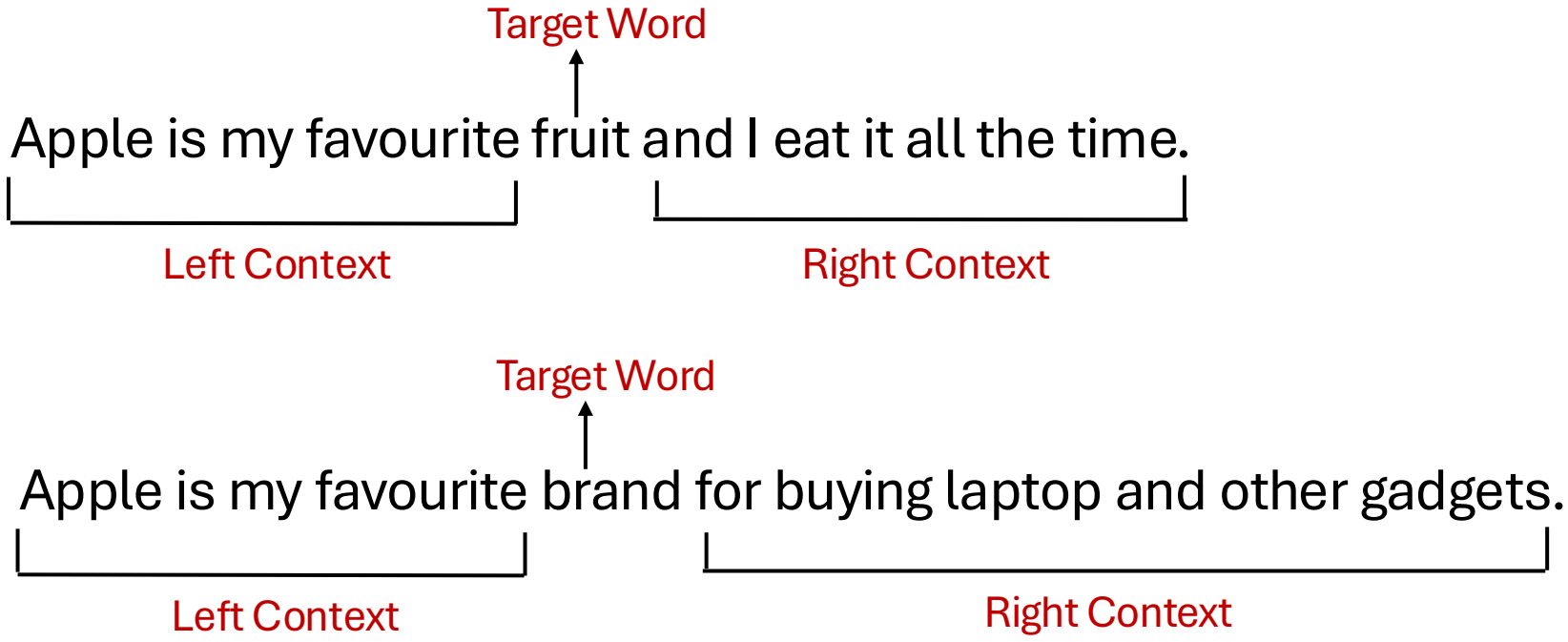
BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin **Ming-Wei Chang** **Kenton Lee** **Kristina Toutanova**
Google AI Language
{jacobdevlin,mingweichang,kentonl,kristout}@google.com

Slides are adopted from Jacob Devlin

Background - Bidirectional Context

- Bidirectional context, unlike unidirectional context, takes into account both the left and right contexts.



Motivation

- **Problem with previous methods:**
  - Language models only use left context or right context.
  - But language understanding is **bidirectional**.
- **Possible Issue:**
  - Directionality is needed to generate a well-formed probability distribution.
  - Words can see themselves in a bidirectional model.

Masked Language Modelling

- Mask out k% of the input words, and then predict the masked words (Usually k = 15%). Example :

I like going to the [MASK] in the evening

↓

park

  - Too little masking: Too expensive to train
  - Too much masking: Not enough context
- The model needs to predict 15% of the words, but we don't replace with [MASK] 100% of the time. Instead:
  - 80% of the time, **replace with [MASK]**
    - Example : like going to the park → like going to the [MASK]
  - 10% of the time, **replace random word**
    - Example : like going to the park → like going to the store
  - 10% of the time, **keep same**
    - Example : like going to the park → like going to the park

Next Sentence Prediction

- To learn relationships between sentences, predict whether Sentence B is actual sentence that proceeds Sentence A, or a random sentence.

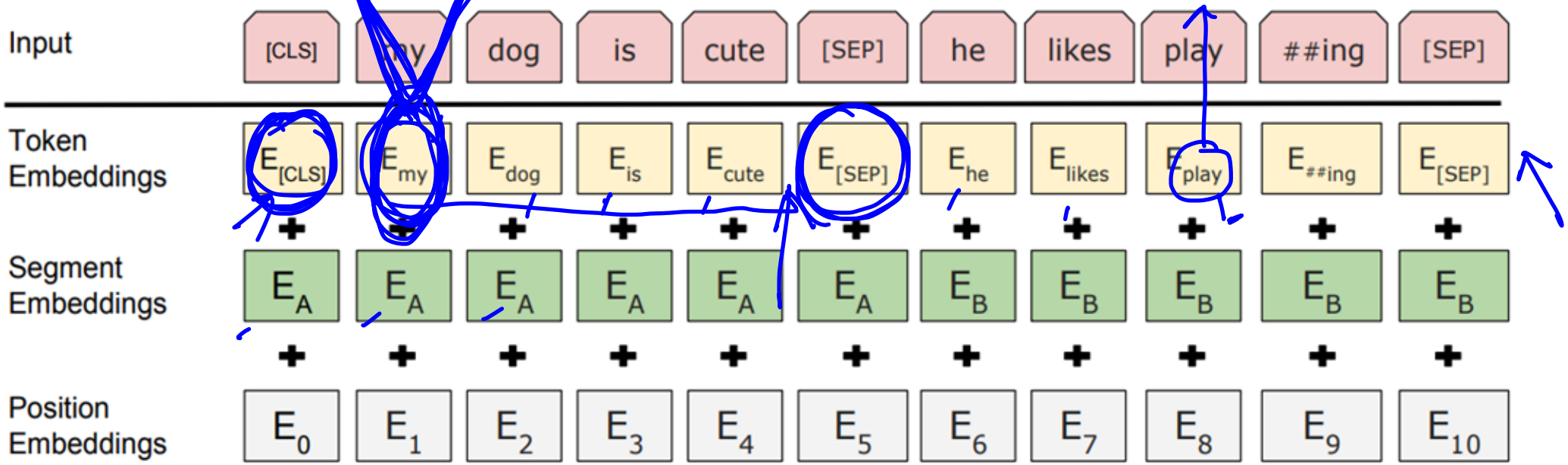
Input = [CLS] I enjoy read [MASK] book ##s [SEP]
I finish ##ed a [MASK] novel [SEP]
Label = IsNext

Input = [CLS] I enjoy read ##ing book [MASK] [SEP]
The dog ran [MASK] the street [SEP]
Label = NotNext

- Important for many important downstream tasks such as Question Answering (QA) and Natural Language Inference (NLI)
- How to choose sentences A and B for pretraining?
  - 50% of the time B is the actual next sentence that follows A (labeled as IsNext)
  - 50% of the time it is a random sentence from the corpus (labeled as NotNext)

Input Representation

- Use 30,000 WordPiece vocabulary on input.
- For a given token, its input representation is constructed by summing the token embeddings, the segmentation embeddings and the position embeddings.



Source of Image : BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (Devlin et al., NAACL 2019)

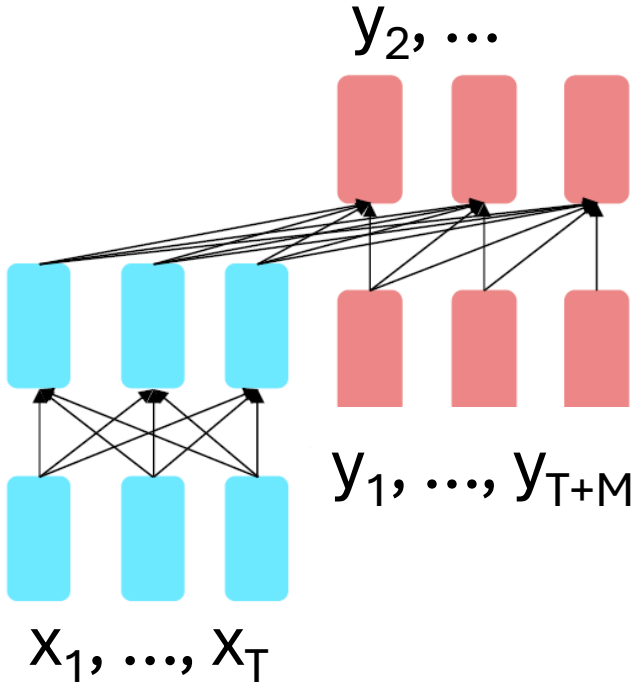
Pre-Training Encoder-Decoder Models

- For **encoder-decoders**, we could do something like **language modeling**, but where a prefix of every input is provided to the encoder and is not predicted.

$$h_1, \ldots, h_T = \textit{Encoder}\,(x_1, \ldots, x_T)$$

$$h_{T+1}, \ldots, h_{T+M} = \textit{Decoder}\,(y_1, \ldots, y_{i-1}, h_1, \ldots, h_T)$$

$$P(y_i \mid y_{<i}, h_{1:T}) = \textit{Softmax}(Wh_i + b)$$



The **encoder** portion benefits from bidirectional context; the **decoder** portion is used to train the whole model through language modeling.

Pre-training

↓

Fine tune

Pre-train → Fine tune → Predict

Post 2017.

2018 - BERT

GPT

⋮

2018

↓

2020

GPT2 - 1.5B

↓ 100x

GPT3 175B

2020

Pre-train → Prompt → Predict

Q-A task | Closed book QA + Extractive QA

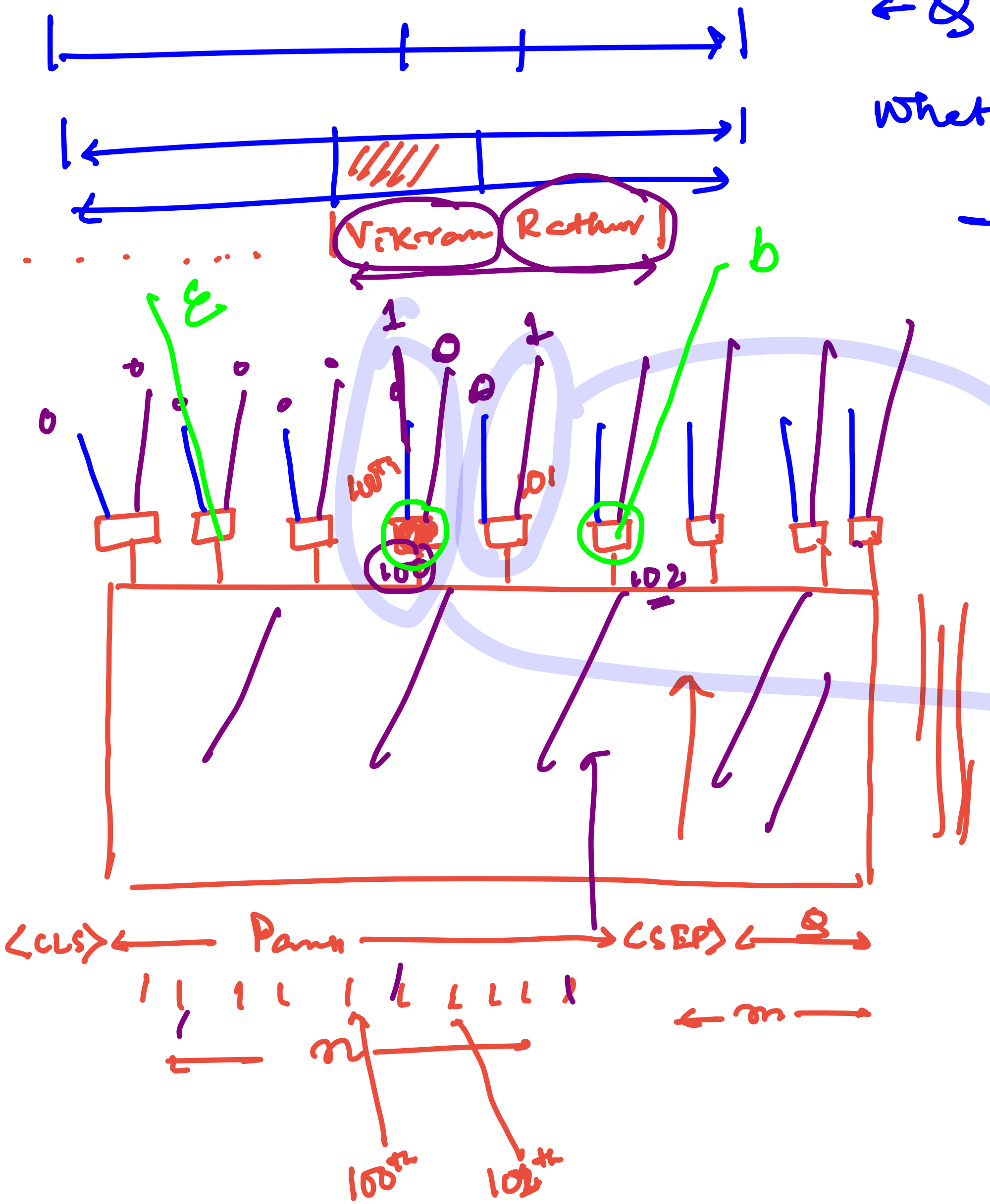
BERT

~~HotpotQA~~

SRK

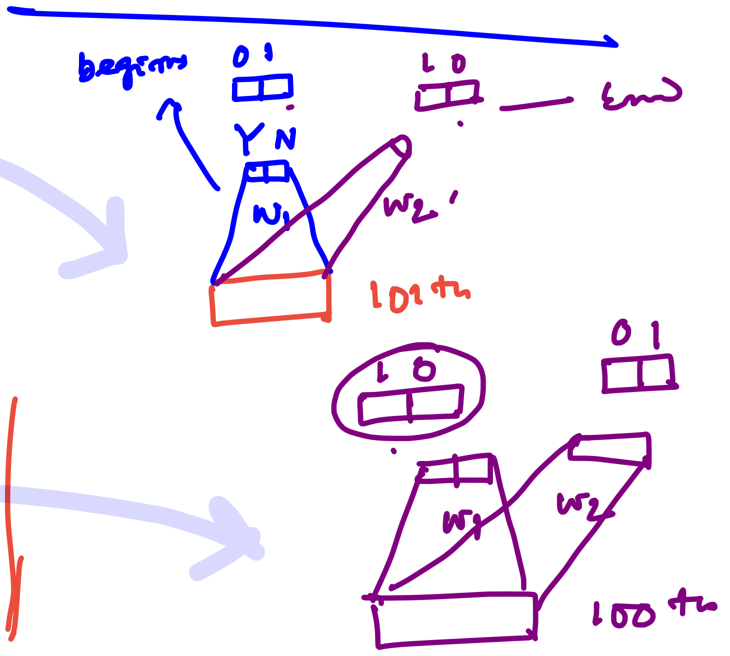
BI

1)

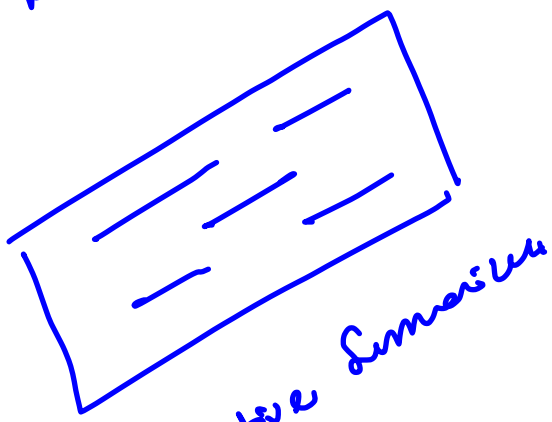


← Q →

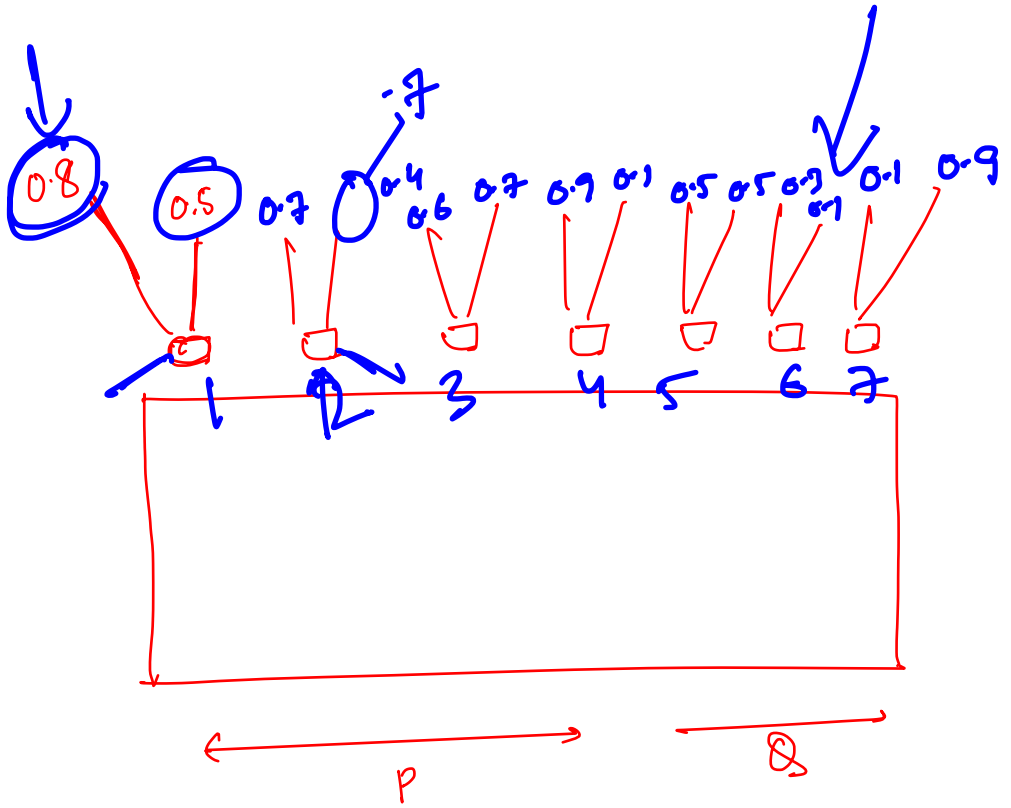
What specific Chart did SRK play in Jawan?



Ex 1



Abstractive Summarization



$p_b \times p_a$

(2, 7)

0.7 X 0.9

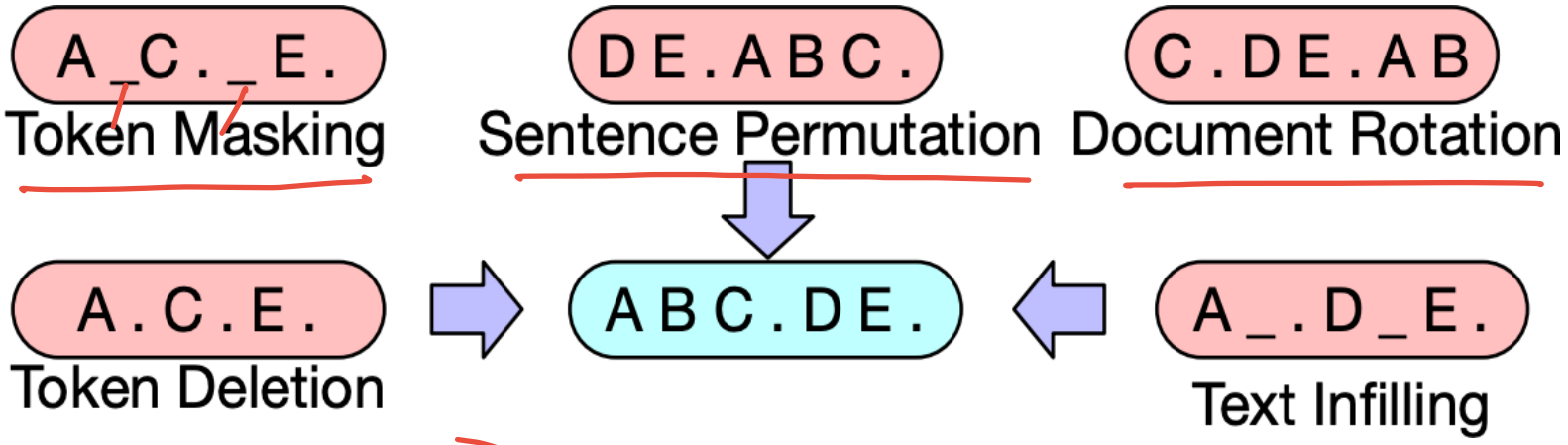
(3, 5)

0.6, X 0.5

Pre-Training Encoder-Decoder Models

- How can we pre-train a model for P(**y** | **x**)?
- **Requirements:**
    1. should use unlabeled data
    2. should force a model to attend from **y** back to **x**

Pre-Training BART (Bidirectional and Auto-Regressive Transformers)

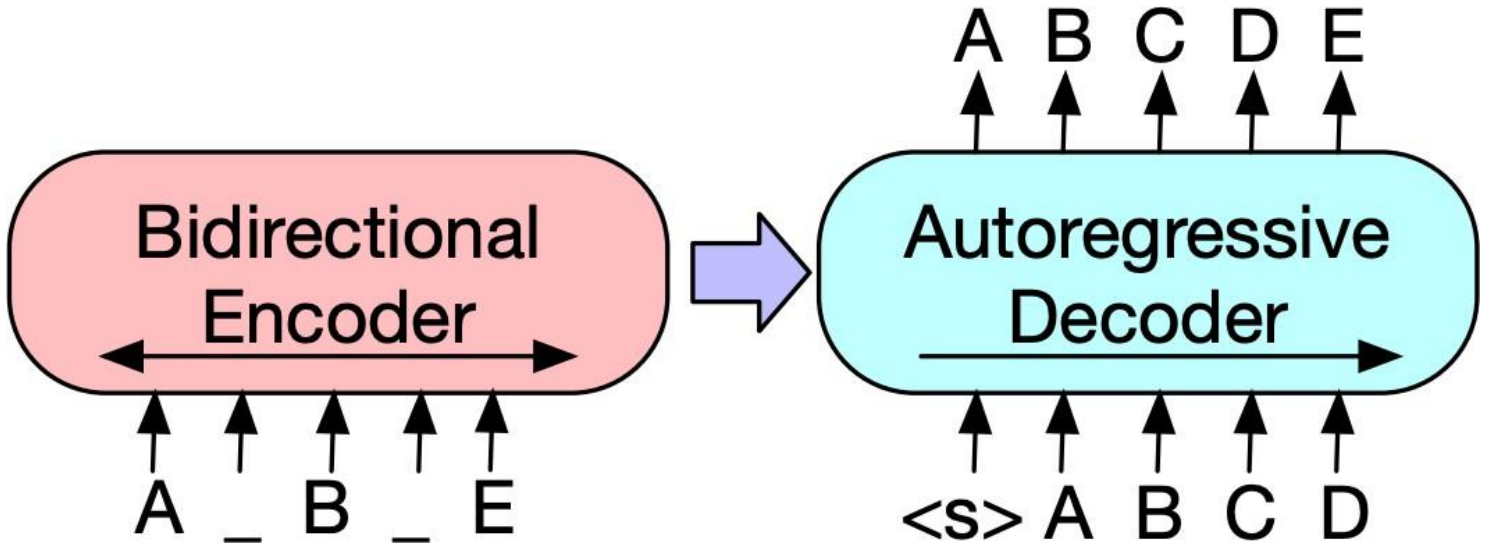


- Several possible strategies for corrupting a sequence are explored in the BART paper.

Lewis et al. (2019), "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension"

Pre-Training BART

- Sequence-to-sequence Transformer trained on this data: permute/make/delete tokens, then predict full sequence autoregressively.



Lewis et al. (2019), "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension"

BART

GPT-3 2020

175 B

"Prompting"

2020

2022

InstructGPT

RLHF

* RL

ChatGPT

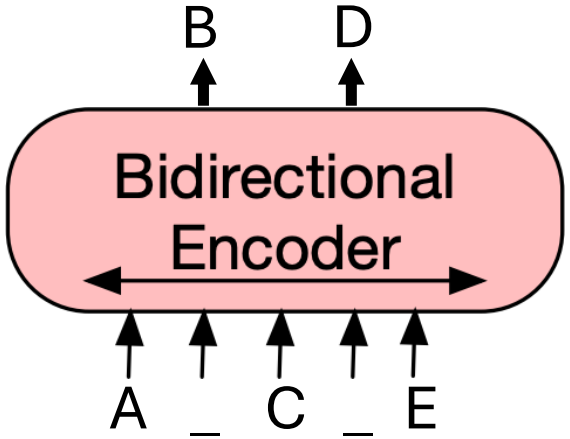
— Dec 2020

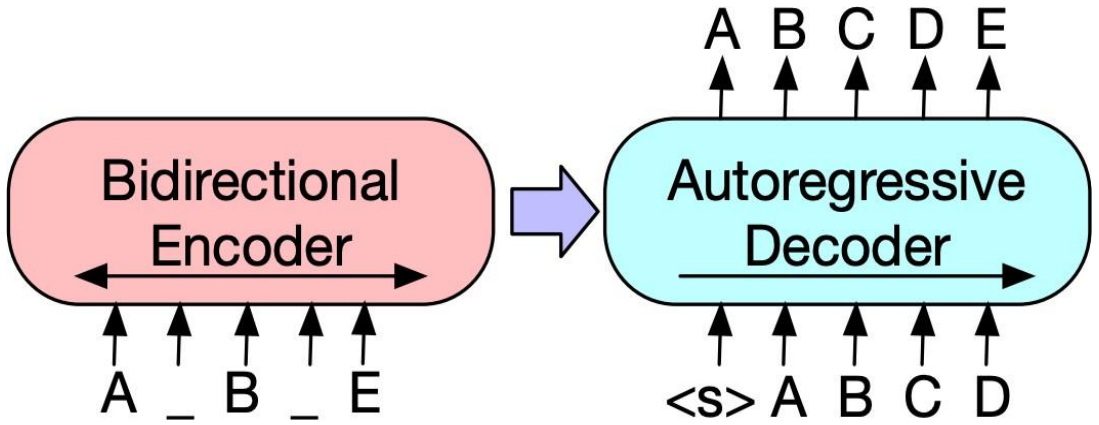
Reinforcement Learning From/with Human Feedback

Alignment

BERT vs. BART

- **BERT:** only an encoder, trained with masked language modeling objective. Cannot generate text or do Seq2Seq tasks (in standard form).





- **BART:** consists of both an encoder and a decoder. Can also use just the encoder wherever we would use BERT.

BART for Summarization

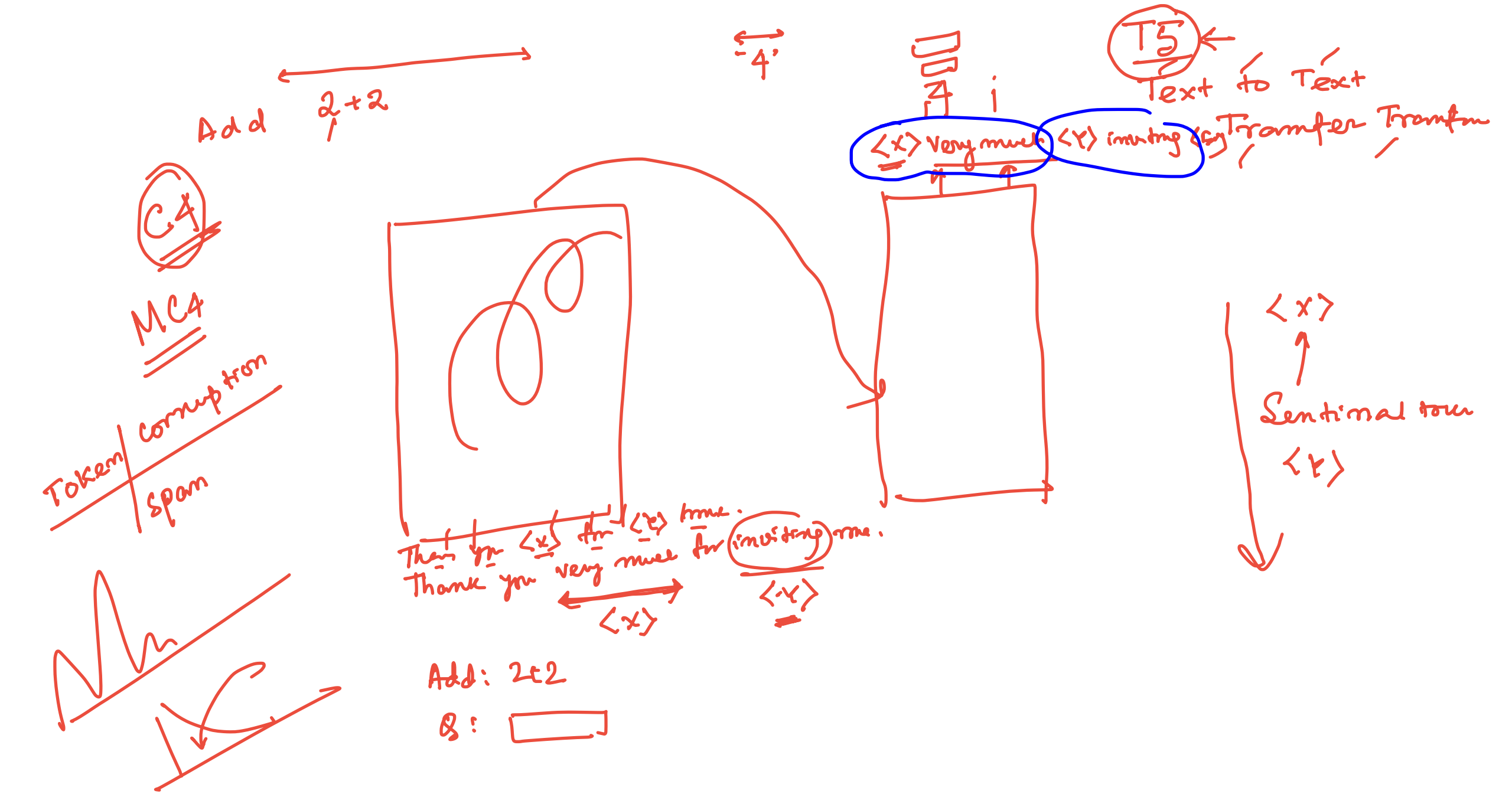
- **Pre-train** on the BART task: take random chunks of text, noise them according to the schemes described, and try to “decode” the clean text
- **Fine-tune** on a summarization dataset: a news article is the input and a summary of that article is the output (usually 1-3 sentences depending on the dataset)
- Can achieve good results even with **few summaries to fine-tune on**, compared to basic seq2seq models which require 100k+ examples to do well

BART for Summarization: Output Example

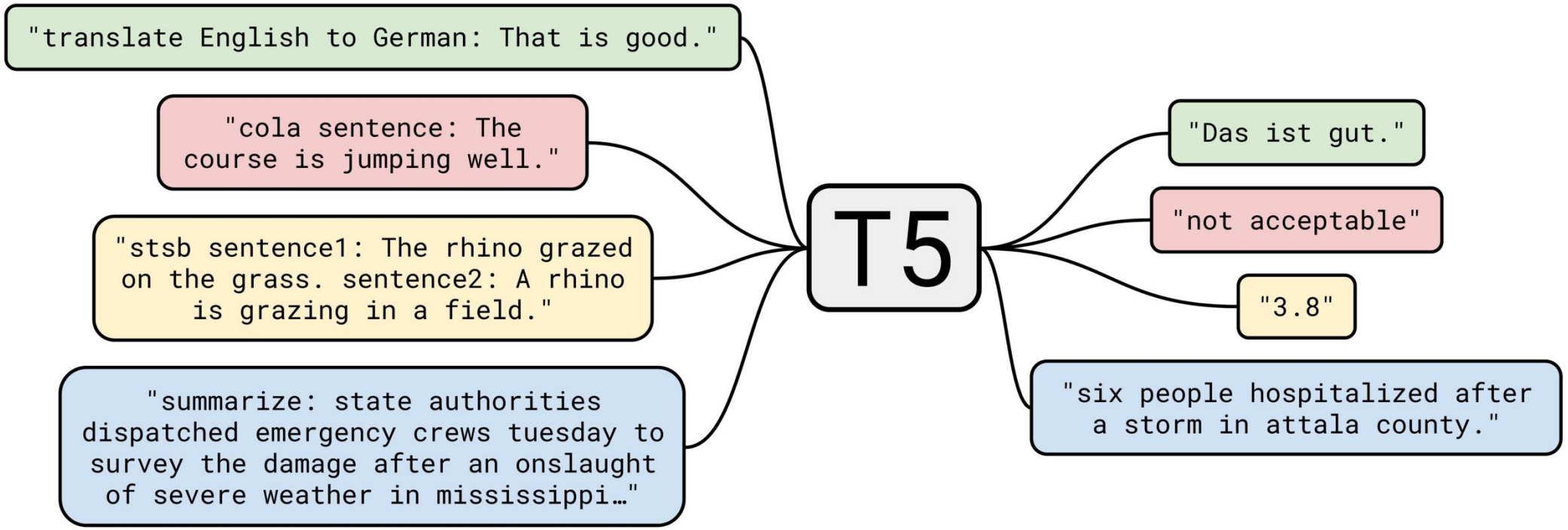
PG&E stated it scheduled the blackouts in response to forecasts for high winds amid dry conditions. The aim is to reduce the risk of wildfires. Nearly 800 thousand customers were scheduled to be affected by the shutoffs which were expected to last through at least midday tomorrow.

Power has been turned off to millions of customers in California as part of a power shutoff plan.

Lewis et al. (2019)



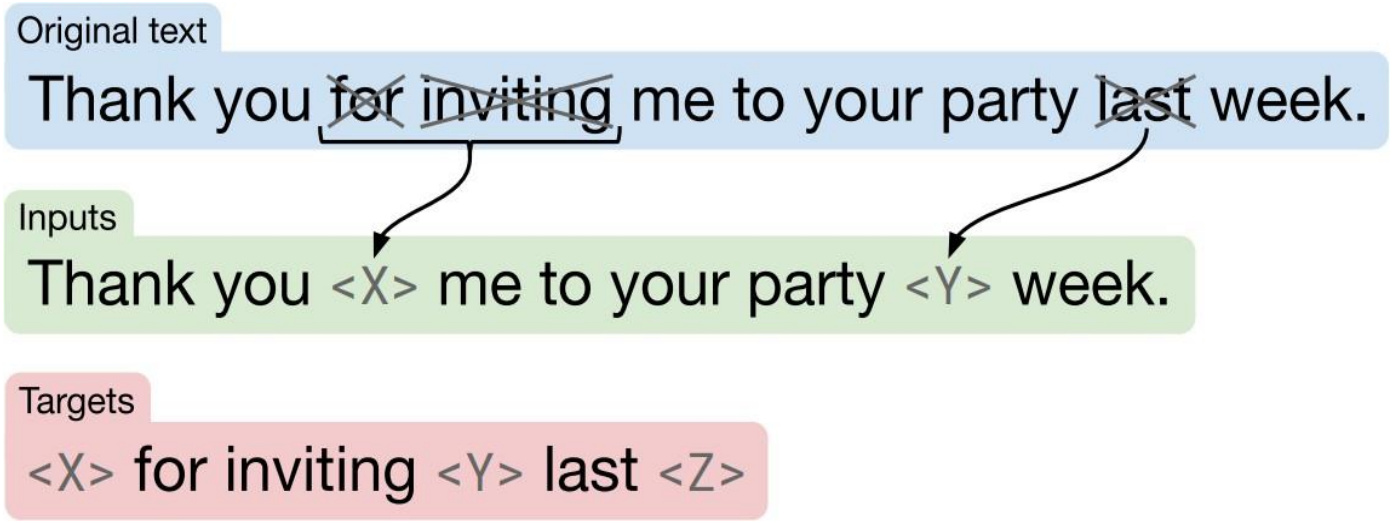
T5: Text-to-Text Transfer Transformer



Raffel et al. (2019), "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer"

Pre-Training T5

- **Pre-training:** similar denoising scheme to BART (they were released within a week of each other in fall 2019)
- **Input:** text with gaps ; **Output:** a series of phrases to fill those gaps.



Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!

Raffel et al. (2019)

Pre-Training Decoder-only Models

GPT and Llama

Recall: Probabilistic Language Models

$$P(X)$$

Sentence/Document

A generative model that calculates the probability of language

Auto-regressive Language Models

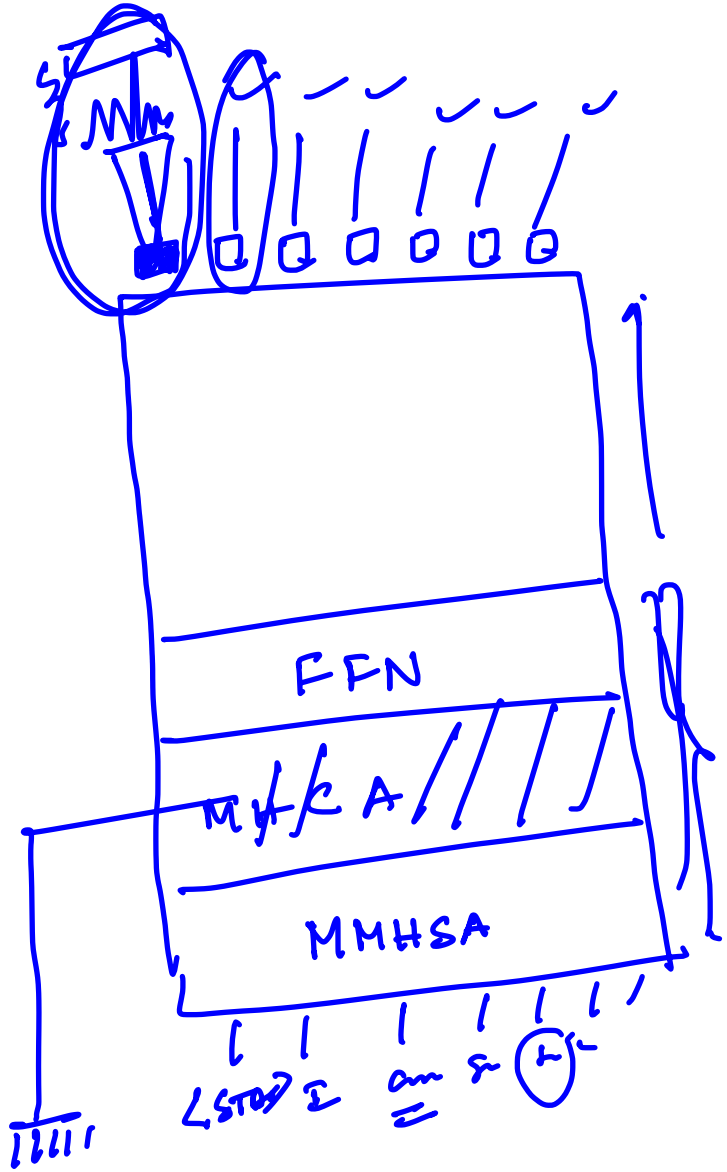
$$P(X) = \prod_{i=1}^{I} P(x_i \mid x_1, \ldots, x_{i-1})$$

Next Token

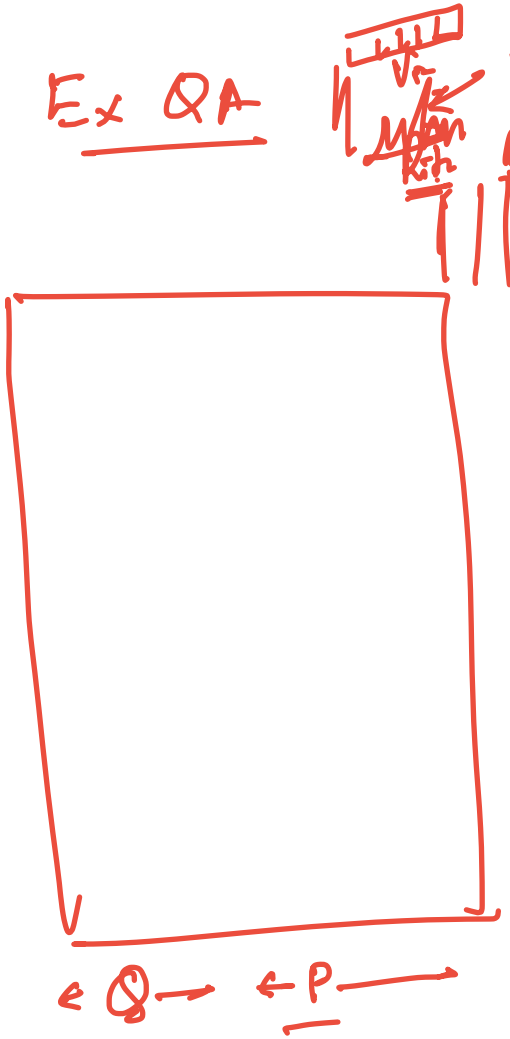
Context

Pretrain
Decode only
CLPT

Finetune ft



Ex QA



Next Token Prediction

- This is essentially **classification**!
  - We can think of neural language models as neural classifiers. They classify prefix of a text into |V| classes, where the classes are vocabulary tokens.

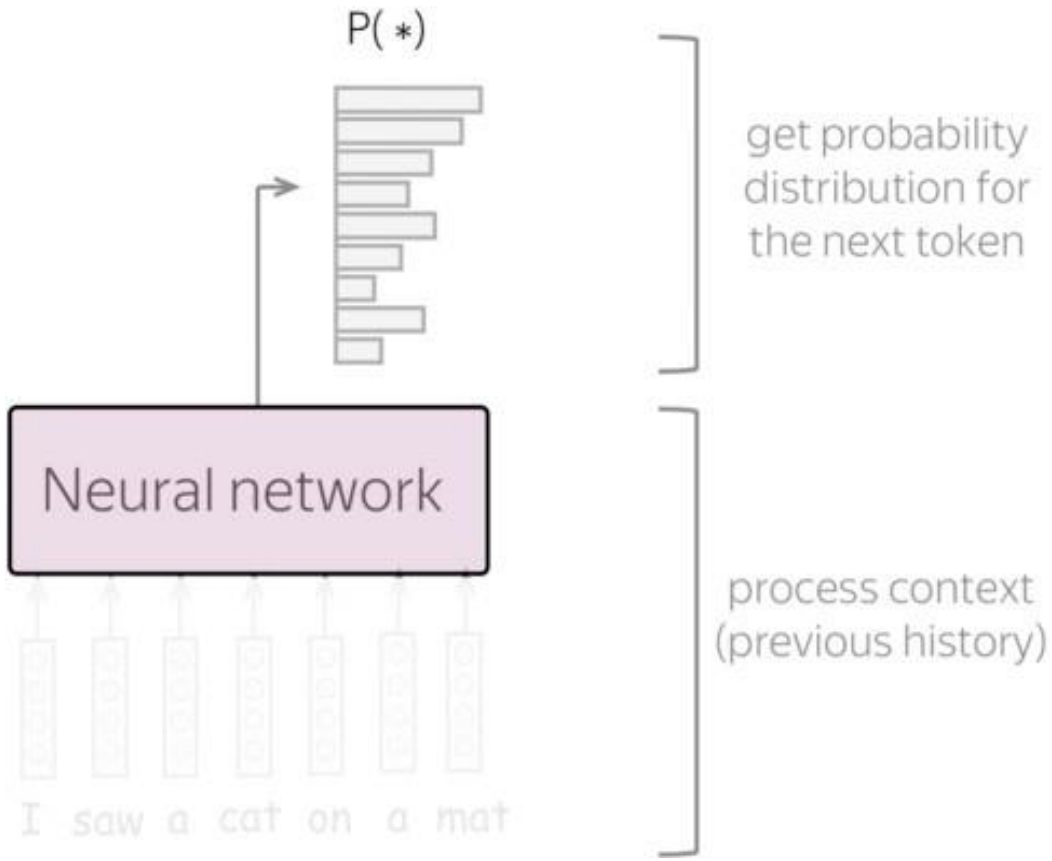
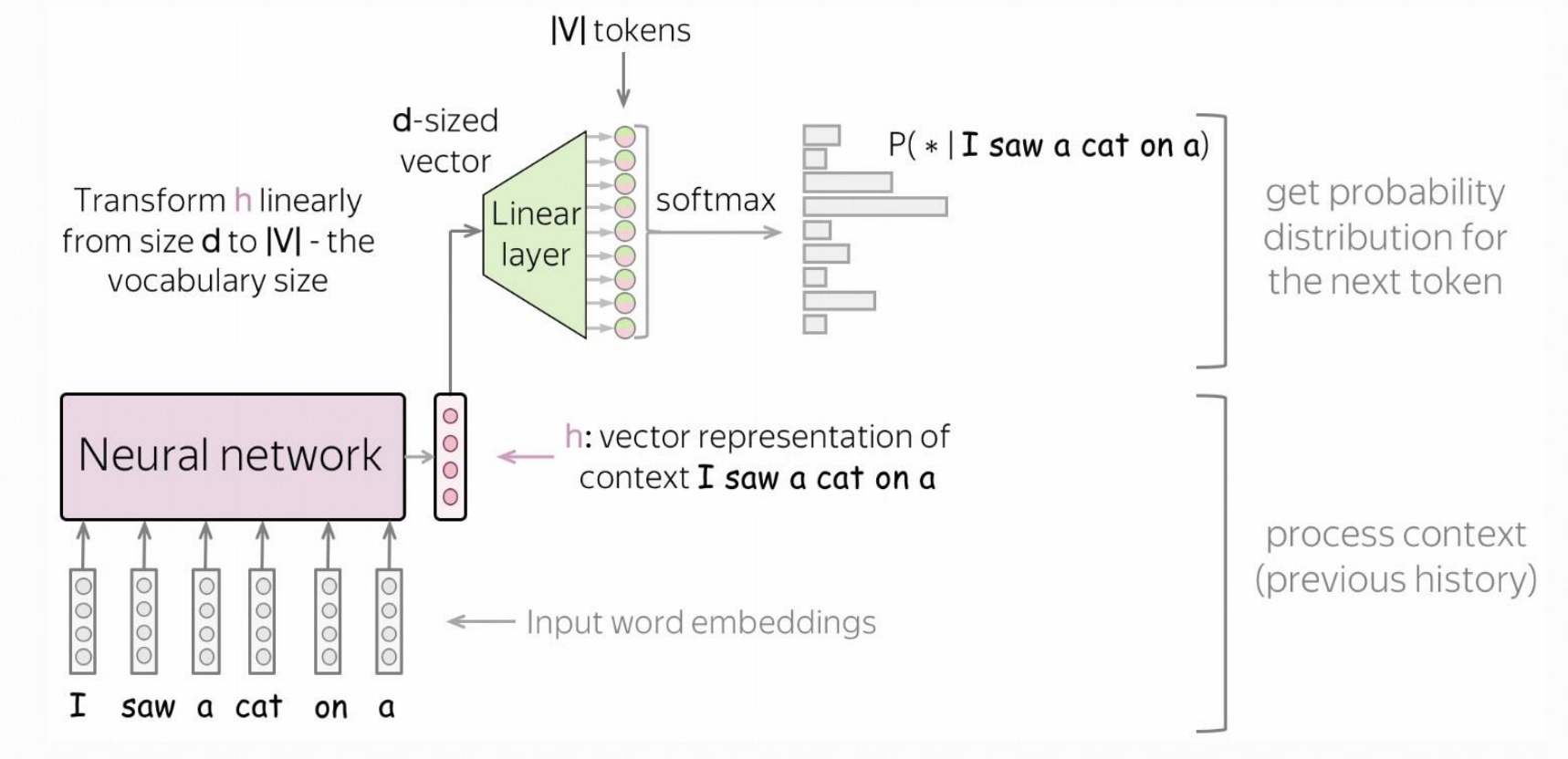


Image Credit: https://lena-voita.github.io/nlp_course/language_modeling.html

Next Token Prediction



- Feed word embedding for previous (context) words into a network.
- Get vector representation of context from the network.
- From this vector representation, predict a probability distribution for the next token.

Image Credit: https://lena-voita.github.io/nlp_course/language_modeling.html